Election Intelligence & Analytics Command Center

Product Requirements Document & Project Proposal — v1.0 (for initial client approval discussion)

[COMPANY NAME] — AI + Data + Digital Intelligence Partner

27 September 2026

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Document Control

Item	Detail
Document	Election Intelligence & Analytics Command Center — PRD & Project Proposal
Version	1.0 — Draft for initial enterprise/client approval discussion
Date	27 September 2026
Prepared by	[COMPANY NAME] (the “Intelligence Partner”)
Prepared for	Leadership of [AGENCY NAME], a political communications / PR agency (the “Agency”), and its campaign client(s) (the “Campaign”)
Classification	Confidential — Commercial in Confidence
Companion artefacts	Executive presentation (35 slides, .pptx + .pdf); synthetic demo dataset generator and charts (/demo); conceptual schema (/poc/schema.sql); API sketch (/poc/openapi.yaml)
Status of legal content	Research summary as of September 2026 — **requires review by the Campaign’s authorised legal/compliance team**

Important notices

1. **This is an analytics/technology proposal, not legal advice.** Final campaign compliance should be reviewed by the Campaign’s authorised legal/compliance team. Regulatory items below are cited to official or reputable sources with dates; anything marked **[VERIFY]** must be confirmed against the primary Election Commission of India (ECI) / Government of India document before it is relied on.
2. **Scope of ethics.** The platform is designed for lawful, privacy-aware, aggregate analytics and operational decision-support. It does **not** profile individuals’ political beliefs, does **not** perform individualised persuasion or micro-targeting on sensitive attributes, does **not** track people covertly, and does **not** scrape personal data without authorisation.
3. **Demo data.** Every number shown in dashboards, charts and sample reports in this document is **DEMO DATA — SYNTHETIC**, generated for a fictional constituency (“Aranyapur AC”). No real voter, person, account or constituency data is used.
4. **Branding.** No party logo or symbol is used. Colour choices (saffron, deep navy, white, gold) are a design direction only. Official party or client marks may be inserted only by the Agency with written authorisation from the rights holder; until then all marks are **placeholders**.
5. **Commercials.** All INR figures are **indicative only; the final commercial proposal depends on constituency, election timeline, data availability, staffing, SLA and technology requirements.**

How to read this document

If you are…	Read
Agency / Campaign leadership (15 minutes)	§1 Executive Summary, §2 Business Problem, §28 Commercials, §30 Risks, §31 What we will not promise
Programme & product owners	§3–§5 Objectives/Scope/Personas, §6 Functional Requirements, §18 Command Center UI, §25 POC
Data / technology leads	§8–§17 Data, Architecture, Data Model, Analytics, ML/NLP, AI & RAG, Security, Model Governance
Legal / compliance	§9 Governance, §10 Election Compliance, §16 Security, §17 Responsible AI, Appendix A Sources

1. Executive Summary

1.1 The product in one sentence

The Election Intelligence & Analytics Command Center converts fragmented campaign, digital, survey, media, field and public data into a single, governed intelligence layer that helps campaign leadership understand *what* is changing, *where* it is changing, *why* it may be changing, and *how* campaign operations are performing.

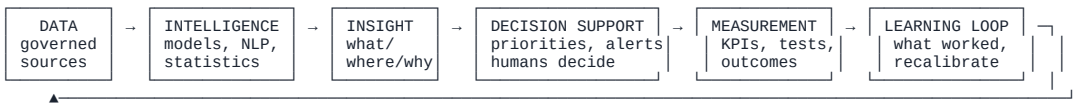
1.2 Why now

Modern campaign organisations generate more data than they can use. Digital teams export platform dashboards; survey vendors send PDFs and SPSS files; media cells clip coverage into WhatsApp groups; field coordinators report in spreadsheets; call-centres run their own tools. Each team reports different numbers on different days. Leadership receives a pile of reports, not intelligence.

Three developments make a unified, governed approach both necessary and possible in 2026:

1. **Regulatory tightening on digital campaigning and AI content.** The ECI's 20 March 2026 framework for the 2026 Assembly elections required MCMC pre-certification for political advertisements on electronic media *including social media*, disclosure of authentic social-media accounts, and full reporting of internet/social-media expenditure; ECI advisories (24 October 2025; reiterated April 2026) require labelling of AI-generated/synthetic campaign content and takedown of unlawful/deepfake content within three hours; MeitY's IT Amendment Rules 2026 (in force 20 February 2026) regulate synthetically generated information. Campaigns now need an **auditable** record of what was published, approved, certified, labelled and spent. *(Sources: Appendix A, L1–L8.)*
2. **Data-protection obligations.** The Digital Personal Data Protection (DPDP) Act 2023 and DPDP Rules 2025 (notified 14 November 2025, phased commencement through 14 May 2027) make purpose limitation, consent, security safeguards, breach reporting and log retention board-level concerns. A platform built in 2026 must be **privacy-by-design** from day one. *(Appendix A, L12–L13.)*
3. **AI is only as good as the data under it.** Large language models can summarise and explain, but without a trusted data layer they fabricate numbers. The value is in **AI on top of verified, lineage-tracked data**, with citations, timestamps and confidence — and humans in the loop.

1.3 The intelligence loop



- **Data** — every source registered, owned, lawful-basis tagged, quality-scored and lineage-tracked.
- **Intelligence** — statistics, forecasting, anomaly detection and multilingual NLP turn raw records into signals *with uncertainty attached*.
- **Insight** — analysts translate signals into "what changed, where, and plausible why", separating facts from interpretation.
- **Decision support** — leadership receives prioritised operational, monitoring and measurement recommendations; **humans decide**.
- **Measurement** — every decision has a KPI; content and operations are measured against baselines.
- **Learning loop** — results feed back into models, KPIs and playbooks; the system gets better every week.

1.4 What the platform delivers

Capability	What leadership gets	Guardrail
Executive Command Center	One Campaign Intelligence Health view, drill-down to 14 modules	Every tile shows source, refresh time and data-quality status
Digital Performance Scorecard	Reach, engagement, video retention, CTR/CPC/CPM, spend by channel and content type	Platform-provided analytics only; API availability verified per platform
Social Listening & Issue Intelligence	Mention volume, sentiment, topic velocity, emerging-issue alerts	Coverage warnings; social ≠ electorate
Media Intelligence	Multilingual article/broadcast classification, tone, issue	Licensed feeds only for broadcast/paywalled

Capability	What leadership gets	Guardrail
	share, regional coverage	content
Survey Intelligence	Weighted estimates with margins of error, significance tests, wave-over-wave trends	No claim that surveys predict results
Field & Call-centre Operations	Activity completion, coverage, bottlenecks, service levels	Aggregate/operational data; no private communications
GIS Intelligence	Constituency/zone-level maps of activity, coverage, issues, historical results	No individual-level targeting
Forecasting & Alerts	Issue-volume, media-volume, digital and turnout-trend forecasts with intervals; 4-level alert engine	Predictions carry assumptions, validation metrics and limitations
AI Campaign Analyst	Natural-language Q&A over approved data with citations, timestamps and confidence	Answers only from approved sources; refuses to fabricate
Election Compliance Control Tower	Content approval, MCMC certification, AI-label status, spend tracking, audit trail	Supports — does not replace — the legal/compliance team

1.5 Positioning — an intelligence partner, not a staffing vendor

> **“We do not simply provide analysts. We build the intelligence infrastructure that turns fragmented campaign data into a continuously learning decision-support system.”**

Capability pillars: **DATA · AI · ANALYTICS · SOCIAL INTELLIGENCE · MEDIA INTELLIGENCE · DIGITAL MEASUREMENT · FIELD INTELLIGENCE · AUTOMATION · COMMAND CENTER.**

The Agency retains ownership of its data, its relationships and its creative/strategic judgement. The Intelligence Partner provides the platform, the governed data layer, the models, the analysts and the operating rhythm.

1.6 What we ask for

Step	Duration	Indicative investment	Decision gate
Proof of Concept — synthetic + limited authorised real data, 6 modules, AI Analyst, alerts, automated daily report	4–6 weeks	₹18–35 lakh (fixed-fee band)	13 measurable acceptance criteria (§25.4)
Tier selection — Starter / Professional / Enterprise	—	See §28	Scope, geography, SLA
Production rollout — 30/60/90-day roadmap to full Command Center	90 days	Setup + monthly retainer + pass-through	Adoption & quality KPIs (§27)

1.7 What we will not promise

No guaranteed election outcome, no exact vote prediction, no individual political-belief inference, no unauthorised personal-data access, no secret tracking or illegal scraping, no private-message monitoring, no guaranteed virality, persuasion or ROI. (Full list: §31.)

1.8 Additional capabilities added after critical self-review

A structured red-team review (§34) added the following beyond the original brief — each is explained where it appears:

1. **Source & Lawful-Basis Registry** with per-source “permitted use” tags enforced at query time (§9).
2. **Metric Store / semantic layer** so every team and the AI use one definition per KPI (§11).
3. **Coverage & Representativeness Index** shown on every social/media tile (§13.2, §18).
4. **Human-labelled multilingual gold sets** (Hindi, English, Hinglish, regional, romanised) with inter-annotator agreement targets (§14.2).
5. **Experiment & test registry** for pre-registered content/format tests with proper statistics (§21.3).
6. **Silence-period & exit-poll calendar guard** that blocks scheduled publishing workflows and restricts poll-related outputs in legally restricted windows (§10).
7. **AI Content Provenance Ledger** (creator, tool, timestamp, label, approval) aligned with the ECI record-keeping advisory (§10).
8. **India-boundary map compliance** — official boundaries only for any national/state outline (§22).
9. **Escalation matrix, SLAs and business-continuity runbooks** for poll-week operations (§24.5–§24.7).
10. **Cost guardrails** (per-module cloud/API budgets, LLM token caps, pass-through transparency) (§28.6).

11. **Red-team prompt suite & hallucination test harness** for the AI Analyst (§15.5).
12. **Exit & data-return plan** — full data export and deletion certificate at contract end (§9).

2. Business Problem

2.1 Why traditional campaign reporting is insufficient

Traditional campaign reporting answers “what did each team do yesterday?” It rarely answers “what changed across the whole campaign, where, and what should we look at next?” The failure is structural, not a matter of effort:

- **Reports are team-shaped, not question-shaped.** Digital reports platform metrics, the media cell reports clippings, the survey vendor reports toplines, field reports activity counts. Nobody owns the cross-source question.
- **Numbers disagree.** “Reach” means one thing in Meta, another in YouTube and a third in a vendor deck. Without a metric dictionary, meetings debate numbers instead of decisions.
- **Latency kills relevance.** A manually compiled weekly deck describes a campaign that no longer exists. Issues surface on social and in regional media hours or days before they reach a leadership review.
- **No uncertainty.** A survey that shows 21% vs 18% is presented as a “rise” without a margin of error; a sentiment score from 2,000 posts is treated as public opinion.
- **No audit trail.** Who approved a creative, whether it was MCMC-certified, whether an AI-altered visual was labelled, what was spent where — these live in e-mails and chats, which is untenable under 2026 regulatory expectations.

2.2 How Excel / WhatsApp workflows create blind spots

Workflow symptom	Blind spot it creates	Consequence
Field reports sent as photos / voice notes in groups	Not machine-readable; lost after scroll	Coverage gaps invisible until too late
Multiple versions of the “master” spreadsheet	No single source of truth	Leadership sees conflicting numbers
Platform screenshots pasted into decks	No lineage, no refresh	Cannot trace or re-compute a number
Media clippings forwarded ad hoc	No classification, no trend	Emerging regional stories missed
Survey PDFs from vendors	Raw data, weights, and methodology not retained	Wave-to-wave comparisons unreliable
Credentials shared by message	No access control, no revocation	Security and privacy exposure
Approvals by thumbs-up emoji	No audit trail	Compliance risk on certification and AI labelling

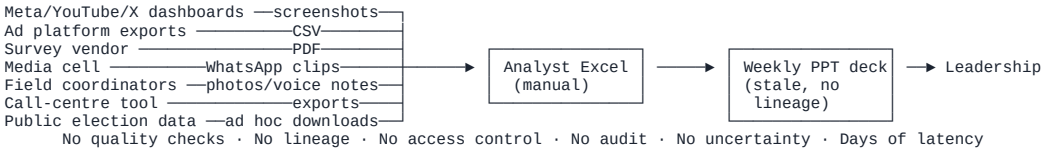
2.3 Problem inventory

#	Problem	Who feels it	Target state
P1	Data scattered across teams	Everyone	Registered sources feeding one governed warehouse
P2	Different teams using different numbers	Leadership	Metric store: one definition per KPI, versioned
P3	Delayed reporting	Leadership, war-room	Daily automated report by 07:00; intraday refresh for priority sources
P4	Manual Excel reporting	Analysts	≥70% of recurring report content auto-generated
P5	No single source of truth	All	Warehouse + semantic layer + lineage
P6	Social data disconnected from survey data	Strategy, research	Common issue taxonomy across social, media, survey, field, call-centre
P7	Field feedback disconnected from digital analytics	Field, digital	Shared geography and time keys
P8	Media monitoring disconnected from campaign performance	Media cell, comms	Media volume/tone alongside digital and field metrics on one timeline
P9	No systematic anomaly detection	War-room	Statistical alert engine with 4 severity levels
P10	No standardised KPI framework	Leadership	3-level KPI tree with leading/lagging/diagnostic/data-quality indicators
P11	Difficulty identifying emerging issues	Research, comms	Topic-velocity and novelty detection over multilingual text
P12	Difficulty comparing time periods	Everyone	Standard period-over-period and rolling-baseline

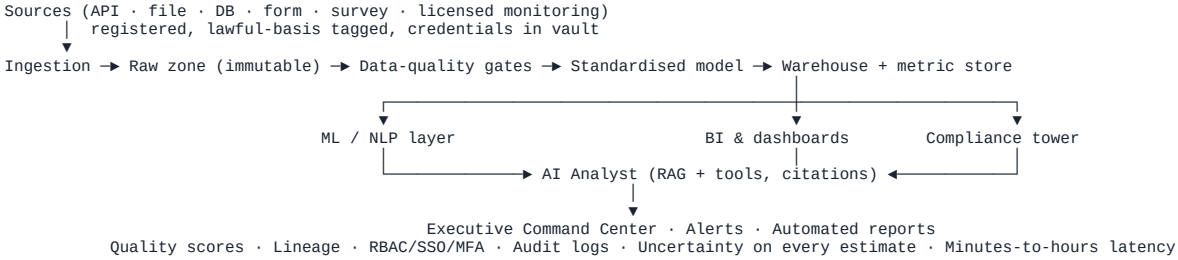
#	Problem	Who feels it	Target state
			comparisons
P13	No confidence/uncertainty in survey interpretation	Leadership	Weighted estimates with MoE and significance flags by default
P14	Poor data quality	Analysts	Automated tests; quality score per source on every dashboard
P15	Lack of auditability	Compliance	Immutable audit log; compliance control tower
P16	Slow decision cycles	Leadership	Decision-support briefs with owners and follow-ups
P17	AI outputs without verified source data	Everyone	Grounded AI Analyst with citations; refuses unsupported answers

2.4 Current state vs future state

Current state (typical):



Future state (Command Center):



3. Objectives & Success Measures

Objective	Measure	POC target	Production target (by day 90)
O1 Single source of truth	% of leadership KPIs served from the metric store	60%	95%
O2 Faster reporting	Time to produce daily war-room report	≤ 30 min analyst review of auto-draft	Auto-draft by 07:00, published by 08:00
O3 Less manual work	% of recurring report sections auto-generated	≥ 50%	≥ 70%
O4 Faster issue detection	Median lag between emerging-issue onset and alert (in back-tests)	≤ 6 h	≤ 2 h for priority sources
O5 Better data quality	Sources with automated quality tests	100% of POC sources	100% of production sources
O6 Trustworthy AI	AI Analyst answers with valid citations on the evaluation set	≥ 95%	≥ 98%
O7 Auditable compliance	Published paid/AI content with complete approval + certification + label record	100% (demo)	100%
O8 Adoption	Weekly active leadership/manager users	Demo	≥ 80% of named users

4. Scope & Non-scope

4.1 In scope

- Ingestion, storage, quality control and modelling of: public/official datasets; historical election results; aggregate demographic and development indicators; campaign-owned first-party data **with documented authorisation**;

consented survey data; platform-provided analytics via permitted APIs/exports; licensed media monitoring; permitted public social data; field reports; lawfully collected call-centre operational data; website analytics; campaign spend and compliance records.

- Descriptive, diagnostic, predictive (with uncertainty) and decision-support analytics at **aggregate** levels (constituency, block, ward/zone, channel, content, issue, time).
- Multilingual NLP for public/licensed text (topic, sentiment, entities, issues, duplicates, toxicity, misinformation-*risk* flags for human review).
- AI Analyst with RAG over approved documents and tool-based access to governed data.
- Command Center dashboards, alert engine, automated daily/weekly reports.
- Election Compliance Control Tower (workflow and audit support).
- Security, privacy, data governance, model governance, and responsible-AI controls.
- Team, operating model, POC, and roadmap.

4.2 Explicitly out of scope (non-scope)

Out of scope	Reason
Individual-level political-belief, ideology or vote-preference inference or scoring	Ethically unacceptable; sensitive; not required for aggregate decision-support
Individualised persuasion, psychological manipulation, micro-targeting on sensitive attributes	Manipulative; conflicts with platform policies and responsible-AI principles
Covert tracking of people, private-message/closed-group monitoring, device tracking	Unlawful/unethical; violates privacy and platform terms
Scraping personal data or platforms in breach of terms of service	Legal and platform risk
Use of voter rolls to build individual profiles	Not required; privacy and legal risk. Aggregate public statistics (e.g., elector counts per AC from ECI statistical reports) are in scope
Generating deceptive content, deepfakes, impersonation, or unlabelled synthetic media	Prohibited by ECI advisories and IT Rules; the platform *tracks labels*, it does not generate campaign creative
Voter suppression, intimidation or discouraging participation	Unlawful and contrary to democratic norms
Publishing exit polls / election-matter in restricted windows	RP Act ss.126, 126A — platform enforces calendar guards
Final legal determinations	Owned by the Campaign's legal/compliance team

5. Users & Personas

Persona	Role	Primary questions	Key screens	Access level
Campaign Director ("Leadership")	Final decision-maker	What changed? What needs my attention? Are operations on track?	Executive Overview, Alerts, Daily Report, AI Analyst	Read all aggregates; no raw PII
Agency Account Lead	Runs the client relationship	Is the programme delivering? Are we compliant?	Executive Overview, Compliance, Reports	Read aggregates; approve reports
Head of Digital	Owns channels and content	Which formats/themes perform? Spend efficiency?	Digital, Content Matrix, Experiments	Digital data incl. spend
Media Cell Lead	Monitors coverage	What's the coverage trend, tone, and emerging stories?	Media Intelligence, Issue Intelligence	Media data
Research / Survey Lead	Owns surveys	Are changes significant? Is the sample healthy?	Survey Intelligence	Survey microdata (pseudonymised)
Field Operations Head	Runs field teams	Where are gaps and bottlenecks?	Field Ops, Geographic	Field data at zone level
Call-centre Manager	Runs helpline	Service levels, issue categories, backlog	Call-centre	Call operational data (no content recordings unless lawfully authorised)
Compliance Officer	Legal/ compliance liaison	Approval, certification, AI labels, spend, silence period	Compliance Control Tower, Audit	Compliance records, audit logs

Persona	Role	Primary questions	Key screens	Access level
Data Analyst / Scientist	Builds analysis	Is the data right? What explains the change?	Data Quality, notebooks, model registry	Governed warehouse (role-scoped)
Platform Admin / Security Lead	Runs platform	Who accessed what? Are pipelines healthy?	Admin, Audit, Pipeline status	Admin; no default data access

6. Functional Requirements

Priority: **M** = Must (POC/MVP), **S** = Should (by day 60), **C** = Could (by day 90+).

6.1 Data ingestion & management

ID	Requirement	Priority
FR-01	Register every data source in a Source Registry with owner, lawful basis/authorisation, permitted purposes, refresh cadence, sensitivity, retention	M
FR-02	Ingest via API connectors (only where access is verified), scheduled file drops (CSV/XLSX/JSON), database replication, manual upload with validation, survey platform exports, form submissions	M
FR-03	Store raw data immutably with ingestion timestamp, source ID and checksum	M
FR-04	Apply automated data-quality tests (schema, completeness, freshness, uniqueness, ranges, referential integrity, anomaly on volumes) and block promotion on critical failures	M
FR-05	Standardise geography (state → district → AC → block/ward/zone), time (IST), language and issue taxonomy	M
FR-06	Maintain column-level lineage from source to dashboard tile and AI answer	S
FR-07	Support pseudonymisation / tokenisation of any personal identifiers at ingestion; drop fields not needed for the stated purpose	M
FR-08	Support deletion requests and retention-based purges with deletion certificates	S

6.2 Analytics & intelligence

ID	Requirement	Priority
FR-10	KPI metric store with versioned definitions consumed by BI, reports and AI	M
FR-11	Period-over-period comparisons (D/D, W/W, vs 4-week baseline) with significance/interval where applicable	M
FR-12	Diagnostic decomposition (“which channel/content/geography/issue drove the change”)	S
FR-13	Survey engine: weighting (post-stratification/raking), design effect, MoE, CIs, cross-tabs, significance testing, quality flags	M
FR-14	Forecasts (issue volume, media volume, digital performance, turnout-trend at aggregate level) with prediction intervals, training window, validation metrics	S
FR-15	Anomaly detection on all priority time series	M
FR-16	Content Performance Matrix and experiment registry	S
FR-17	Field and call-centre operational analytics	M
FR-18	GIS layers at aggregate levels with official boundaries where available	S

6.3 NLP & AI

ID	Requirement	Priority
FR-20	Language identification incl. Hinglish and romanised Indian languages	M
FR-21	Topic/issue classification against the campaign issue taxonomy	M
FR-22	Sentiment/tone classification with confidence; validated against human-labelled test sets per language	M
FR-23	Named-entity recognition for public figures, organisations, places, schemes	S
FR-24	Near-duplicate/coordinated-copy detection	S
FR-25	Toxicity/safety and misinformation-*risk* flags routed to human review (never auto-labelled as “fake”)	S

ID	Requirement	Priority
FR-26	AI Analyst: natural-language questions answered via tools (SQL/metrics/search/vector) with citations, timestamps and confidence	M
FR-27	RAG over approved document library with metadata filters and access control	M
FR-28	AI Analyst refuses questions outside policy (individual profiling, persuasion scripts, outcome guarantees) and explains why	M

6.4 Command Center, alerts & reporting

ID	Requirement	Priority
FR-30	Campaign Intelligence Health Score with drill-down to 8 Level-2 domains	M
FR-31	14 dashboard modules (see §18)	M (6 in POC) / S (rest)
FR-32	Alert engine with Critical/High/Medium/Low levels, routing, acknowledgement and closure notes	M
FR-33	Automated Daily War-Room Report (auto-draft, analyst review, approval, distribution)	M
FR-34	Weekly Executive Report (10–15 pages)	S
FR-35	Export to PDF/PPTX with source footnotes and “data as of” timestamps	S

6.5 Compliance Control Tower

ID	Requirement	Priority
FR-40	Track each paid or AI-assisted content item: owner, approver, approval time, MCMC certification ref/status, platform, spend, AI-label status, takedown events	M
FR-41	Silence-period and exit-poll calendar per phase/constituency; blocks scheduled report types and publishing workflows in restricted windows	M
FR-42	Spend tracking vs budget and vs statutory limits (as configured by the compliance team) with alerts	S
FR-43	Immutable audit trail for approvals, overrides and data access	M

6.6 Administration & security

ID	Requirement	Priority
FR-50	SSO (SAML/OIDC), MFA, RBAC + row/column-level security	M
FR-51	Secrets in a managed vault; API key rotation	M
FR-52	Environment separation (dev/test/prod)	M
FR-53	Admin audit and access reviews	S

7. Non-functional Requirements

Category	Requirement
Availability	99.5% monthly for dashboards (POC: best effort); 99.9% during declared high-intensity phase (Enterprise tier)
Freshness	Priority sources ≤ 60 min (where source permits); daily sources by 06:00 IST
Performance	Executive dashboard P95 load < 3 s; AI Analyst P95 < 15 s for tool-based answers
Scalability	Up to 50 ACs / 1 state in Professional tier; multi-state in Enterprise
Security	Encryption in transit (TLS 1.2+) and at rest (AES-256/KMS); least privilege; MFA mandatory
Privacy	Data minimisation; pseudonymisation of identifiers; DPDP-aligned notices/consent for first-party data
Data residency	Primary storage in Indian cloud regions; CERT-In log retention (180 days in India)
Auditability	Every KPI traceable to source; every AI answer logs prompt, tools called, sources, model version
Accessibility	WCAG 2.1 AA for dashboards; colour-blind-safe palettes validated
Localisation	UI English + Hindi; content analysis in configured Indian languages
Recoverability	RPO ≤ 1 h, RTO ≤ 4 h for production (Professional); RPO ≤ 15 min, RTO ≤ 1 h (Enterprise, high-intensity phase)
Maintainability	Infrastructure-as-code; CI/CD; data tests in CI; documented runbooks
Cost control	Per-module budget alerts; LLM token caps per user/day; monthly cost report

8. Full Data Ecosystem

Principle: a source enters the platform only after it is registered with an owner, a lawful basis/authorisation, permitted purposes and a retention rule (§9). **API availability and permitted fields must be verified against the current platform documentation and access level** — no connector is assumed to exist.

8.1 A. Official / public data

Source	Typical content	Access pattern	Notes / caveats
Election Commission of India — Statistical Reports & Index Cards (eci.gov.in/statistical-reports; ECINET app)	AC/PC-wise electors, polling stations, turnout, gender-wise participation, party vote share, constituency-wise detailed results	Download (PDF/XLS) → parsed & validated	For 2026 Assembly elections ECI published index cards/statistical reports within 72 hours of results (PIB, 2026). Use official files as the system of record
Historical results — curated research datasets (e.g., TCPD Lok Dhaba, Ashoka University)	Cleaned constituency/candidate results from 1962	Download; **licence/terms to be confirmed**	Reconcile against ECI originals; record dataset version
Candidate affidavits / disclosures (public)	Assets, liabilities, education, cases as disclosed	Public ECI portals; manual/structured extraction	Use only as publicly disclosed; no enrichment with private data
Census of India & official statistics	Population, households, amenities (Census 2011); Census 2027 in progress — house-listing Apr–Sep 2026, population enumeration reference date 1 Mar 2027	Download	Census 2011 is dated; plan to update when Census 2027 releases become available
Government reports & development indicators (data.gov.in, NFHS, PLFS, UDISE+, state dashboards)	Health, education, employment, infrastructure indicators	Download / OGD APIs where available	Aggregate only; note survey years and geographic resolution
Constituency / district boundaries	AC/PC/district geometry	Official sources preferred; community datasets (e.g., DataMeet, CC BY 2.5 India) for analysis with attribution	Some community AC boundaries pre-date delimitation — verify; official India outline required for national maps

8.2 B. Campaign-owned data (only where lawfully obtained and authorised)

Source	Examples	Controls
Website analytics	GA4 or equivalent: sessions, sources, landing pages, events	Consent banner/cookie policy; IP anonymisation; no PII in events
Social-media analytics (owned accounts)	Page/channel insights	Admin-authorised tokens; permitted fields only
Ad-platform reporting	Spend, impressions, reach, clicks, CPM/CPC	Platform policy compliance; MCMC certification reference captured per ad
CRM records	Volunteers, supporters who opted in, event registrations	DPDP-aligned notice and consent; purpose-bound; pseudonymised in analytics
Event records	Event type, location, time, estimated attendance	Aggregate; no facial recognition
Volunteer operations data	Tasks, assignments, completion	Operational only; access by role
Campaign activity data	Planned vs completed activities	Operational
Call-centre operational data	Volumes, wait times, categories, resolution	No recordings/transcripts unless lawfully authorised and disclosed; PII minimised
Survey data	See C	Consent recorded; microdata pseudonymised
Feedback forms	Event/website feedback	Consent; free text PII-scrubbed

8.3 C. Survey data

Types: opinion surveys (issue and attitude, **aggregate reporting only**), issue surveys, awareness surveys, satisfaction surveys, event feedback, campaign feedback.

Concept	What it means	How the platform handles it
Sampling	How respondents are selected (probability: SRS, stratified, cluster/multi-stage; non-probability: quota, online panel)	Sampling design stored as metadata; non-probability samples flagged with caveats
Sample size	Number of completed interviews; drives precision	Calculator: $n = z^2 \cdot p(1-p)/e^2 \times deff$; e.g., ±3 pts at 95% with p=0.5 and deff=1.5 → ≈1,600 completes
Weighting	Adjusting sample to known population margins (age, gender, area type, region)	Post-stratification or raking (iterative proportional fitting); weight trimming; weights versioned
Confidence interval	Range likely to contain the population value (e.g., 95%)	Shown on every estimate
Margin of error	Half-width of the CI; increases with weighting (design effect)	MoE uses effective sample size (Kish: $n_{eff} = (\Sigma w)^2/\Sigma w^2$)
Non-response bias	Those who don't respond differ from those who do	Response-rate tracking; comparison to benchmarks; weighting
Response bias	Social desirability, acquiescence, question-order, interviewer effects	Questionnaire review; interviewer-level QC; mode comparison
Questionnaire design	Neutral wording, balanced scales, order rotation, pre-testing	Design checklist; cognitive pre-test; translation back-check
Quality control	Speeders, straight-liners, duplicates, GPS/time anomalies (for tablet F2F), back-checks	Automated flags; exclusion rules documented per wave

8.4 D. Digital data — API reality check

For every platform: API availability and permitted fields must be verified against the current platform documentation and access level. The table records what public documentation indicated as of September 2026; it is a starting point for the discovery phase, not a guarantee.

Platform	Likely access route	Observations (verify)
Meta — Facebook Pages	Graph API page insights (page admin token); Business Suite exports	Access limited to owned/authorised assets; permissions require app review
Meta — Instagram	Instagram Graph API / "Instagram API with Instagram Login" for Business/Creator accounts	Instagram Basic Display API reached end-of-life on 4 Dec 2024; personal accounts not supported
Meta — Ads	Marketing API insights for authorised ad accounts; **Ad Library** (and Ad Library API) for public transparency on ads about social issues, elections or politics	Ad Library API requires identity/country verification. Political advertisers must complete Meta's authorisation and disclaimer process
Meta — public content research	Meta Content Library & API	Restricted to eligible academic/non-profit researchers (via ICPSR); CrowdTangle was shut down on 14 Aug 2024. **Not a general campaign listening tool**
YouTube	YouTube Data API v3 (public metadata), YouTube Analytics / Reporting API (owned channels)	Quotas apply; owned-channel analytics require channel authorisation
X (Twitter)	X API	Since Feb 2026 new developers default to pay-per-use credits; legacy Basic/Pro tiers being migrated (reported Aug 2026). **Budget and read caps must be confirmed**
Google Search / Ads	Google Ads API for owned accounts; Google Ads Transparency Center	Google requires India election-ad advertiser verification and an ECI/authorised **pre-certificate for each election ad**
Google Trends	Web interface; official **Trends API is an application-gated alpha** (announced 24 Jul 2025)	Do not assume access; treat Trends as indicative only
Website analytics	GA4 Data API / exports; Search Console API	Owned properties only
Listening vendors	Licensed social-listening tools	Vendor must document lawful data sourcing and platform compliance; contract includes audit rights

8.5 E. Media data

Source	Access	Notes
News websites & RSS	Public feeds; licensed aggregators	Respect robots.txt and terms; store headline, URL, snippet, metadata; full text only if licensed
Public news feeds / wire	Licensed	Contracted terms
TV / broadcast	Licensed monitoring vendor transcripts/clips	**Only where licensed**; no unlicensed recording/redistribution
Public YouTube news content	YouTube Data API (metadata, public captions where permitted)	Quotas; ToS
Search trends	Google Trends (web/alpha API) where legally available	Relative index, not volume
Regional & hyperlocal media	Licensed vendors / public feeds	Priority for Indian-language coverage

8.6 F. Field data

Daily field reports (structured mobile forms), event attendance (headcount estimates, not identities), campaign activity completion, local issue reports (categorised), operational status, geographic coverage (zone-level), volunteer activity metrics (counts, completion, timeliness). Collected via a structured form app (e.g., ODK/KoboToolbox-class tool or the campaign's own app) with validation, offline support and geotag at *site* level only where operationally necessary and disclosed.

9. Data Governance Framework

9.1 Governance model

Element	Design
Data ownership	Campaign/Agency is the data owner (and, under DPDP, the Data Fiduciary for personal data it determines purposes for); the Intelligence Partner acts as **Data Processor** under a Data Processing Agreement. Each source has a named business owner and technical steward
Source Registry	Mandatory record per source: ID, owner, description, lawful basis/authorisation evidence, permitted purposes, sensitivity class, fields collected, refresh cadence, retention, deletion method, contract/licence ref, API credential ref
Permitted-use tags (added)	Each dataset carries purpose tags (e.g., ops_analytics, media_monitoring, survey_aggregate). Queries/AI tools check tags at run-time; a request outside permitted purposes is refused and logged
Data dictionary	Business definition, type, allowed values, owner, sensitivity per field; published in the catalogue
Metadata & lineage	Technical metadata and column-level lineage captured by the orchestration/transform layer (dbt docs / OpenLineage-compatible)
Access controls	RBAC via SSO groups; row-level security by geography/team; column masking for sensitive fields; just-in-time elevated access with approval
Encryption	TLS in transit; KMS-managed keys at rest; field-level encryption for any personal identifiers retained
Audit logs	Immutable, append-only logs of logins, queries on sensitive tables, exports, AI prompts/answers, approvals; retained ≥ 1 year (DPDP Rules, Rule 6) and ≥ 180 days in India (CERT-In) — whichever is longer
Retention policy	Per source; default: raw social/media 12 months, survey microdata 24 months (pseudonymised), operational logs 13 months, compliance records per legal advice (election-related records typically retained until expiry of election-petition limitation and any litigation) **[VERIFY with legal]**
Deletion policy	Automated purges on expiry; verified deletion from backups on rotation; deletion certificates at contract end
Consent/authorisation tracking	Consent artefacts (notice version, timestamp, channel, purpose) stored for first-party personal data; authorisation letters for platform accounts; licences for media feeds
Data minimisation	Collect only fields needed; drop handles/IDs from public social data after de-duplication

Element	Design
	unless required and permitted; aggregate early
Purpose limitation	Enforced through permitted-use tags and RBAC; new purposes require governance approval
Data-quality monitoring	Tests on freshness, volume, schema, nulls, ranges, duplicates; quality score per source on every dashboard
Incident response	Runbook: detect → triage → contain → notify (CERT-In within 6 hours for reportable incidents; DPDP Board/affected principals per Rule 7 timelines once in force) → recover → post-mortem
Backup/recovery	Daily snapshots + point-in-time recovery for warehouse and app DB; quarterly restore tests
Vendor governance	Due diligence for listening/media/survey/cloud/LLM vendors: data sourcing, security certifications, sub-processors, residency, DPA, audit rights, exit terms
API credential management	Service accounts per connector; secrets in vault; least-privilege scopes; rotation ≤ 90 days; revocation on staff exit
Exit & data return (added)	At contract end: full export in open formats, documentation, and deletion certificate

9.2 Data Governance Checklist

#	Control	Evidence	Owner	POC	Prod
1	Every source registered with owner and lawful basis	Source Registry entry	Data Steward	✓	✓
2	Authorisation evidence for each platform account/API	Signed authorisation / admin grant	Client + Partner	✓	✓
3	Licence terms captured for media/listening/survey vendors	Contract ref	Procurement	✓	✓
4	Permitted-use tags assigned per dataset	Catalogue	Data Steward	✓	✓
5	Data dictionary published for all modelled tables	dbt docs / catalogue	Analytics Lead	✓	✓
6	PII inventory and minimisation review	PII register	Privacy Lead	✓	✓
7	Consent notices reviewed (DPDP-aligned) for first-party collection	Notice versions	Legal	—	✓
8	RBAC matrix approved	Access matrix	Security Lead	✓	✓
9	SSO + MFA enforced	IdP config	Security Lead	✓	✓
10	Encryption at rest and in transit verified	Config evidence	DevOps	✓	✓
11	Secrets in vault; no credentials in code or chat	Scan report	DevOps	✓	✓
12	Audit logging enabled and tamper-evident	Log config	Security Lead	✓	✓
13	Retention and deletion schedules configured	Policy + jobs	Data Steward	✓	✓
14	Data-quality tests on every source	Test report	Data Engineer	✓	✓
15	Lineage available source → tile	Lineage graph	Data Engineer	Partial	✓
16	Incident response runbook tested	Tabletop record	Security Lead	—	✓
17	Backup restore tested	Restore log	DevOps	—	✓
18	Vendor due diligence completed	DD checklist	Procurement	Partial	✓
19	Model cards for every production model	Model registry	DS Lead	✓	✓
20	Quarterly access review	Review record	Security Lead	—	✓
21	Exit/data-return plan agreed	Contract schedule	Programme Director	✓	✓

9.3 DPDP readiness note

The DPDP Rules 2025 were notified on 14 November 2025 with staggered commencement (immediate for administrative provisions including the Data Protection Board; further provisions from 14 November 2026 and 14 May 2027). Rule 6 (reasonable security safeguards, including log retention of at least one year) and Rule 7 (breach intimation, with a detailed report to the Board within 72 hours) are among the operational obligations. **Our design builds to the fully-in-force standard now**, so the platform does not need re-engineering as commencement dates arrive. Exact applicability of each rule to the Campaign on a given date is a legal determination **[VERIFY with legal]**.

10. Election Compliance

> **This is an analytics/technology proposal, not legal advice. Final campaign compliance should be reviewed by the Campaign's authorised legal/compliance team.** The summary below reflects public sources as of September 2026; primary ECI documents were not directly retrievable from our research environment in every case, so items drawn from reputable secondary reporting of ECI press notes are marked **[VERIFY]**.

10.1 Regulatory anchors relevant to the platform

Area	What the source says (summary)	Platform implication	Source (Appendix A)
Model Code of Conduct (MCC) & social media	ECI's 25 Oct 2013 instructions apply the MCC to content posted on the internet/social media by candidates and parties, require pre-certification of political ads on internet-based media, and require inclusion of social-media campaign expenditure (ad payments, content development, staff operating accounts) in expenditure statements	Compliance tower tracks content approvals, certification, and full digital spend categories	L7
Pre-certification of political ads (2026)	ECI press note of 20 Mar 2026 (for Assam, Kerala, Puducherry, Tamil Nadu, West Bengal and bye-elections) required parties, candidates and associated organisations to obtain **MCMC pre-certification** for political ads on electronic media including social media/internet platforms, bulk SMS/voice; candidates to disclose authentic social-media accounts; parties to file expenditure statements within 75 days incl. internet/social-media spend **[VERIFY]**	Certification reference is a **mandatory field** before a paid item can be marked "ready"; alert on missing ref	L1
Print ad pre-certification	ECI (6 Apr 2026) reiterated pre-certification of political advertisements in print media in the pre-poll window **[VERIFY exact window]**	Calendar-aware rule in compliance tower	L2
AI-generated / synthetic content	ECI advisory of 24 Oct 2025: synthetically generated or AI-altered image/audio/video used for campaigning must be labelled ("AI-Generated", "Digitally Enhanced" or "Synthetic Content") covering ≥10% of visible area (audio: during first 10% of duration); disclose the responsible entity in metadata or caption; no content misrepresenting identity/appearance/voice without consent; takedown within 3 hours of notice; **maintain internal records** of AI-generated material (creator, timestamp) **[VERIFY]**. Earlier advisories: 6 May 2024 (deepfakes, 3-hour removal), 16 Jan 2025 (labelling). ECI reiterated in April 2026	**AI Content Provenance Ledger**; label-status field; takedown SLA timer; audit trail	L3–L6
IT Rules — synthetic	MeitY IT (Intermediary Guidelines and	Platform's own publishing partners	L8

Area	What the source says (summary)	Platform implication	Source (Appendix A)
information	Digital Media Ethics Code) Amendment Rules 2026, notified 10 Feb 2026, in force 20 Feb 2026: due-diligence obligations for synthetically generated information (SGI), clear/prominent labelling, metadata/identifiers where feasible, 3-hour takedown for intermediaries on valid notice	(intermediaries) will enforce; campaign content must be labelled consistently	
Silence period	RP Act 1951 s.126: no display of election matter by television or similar apparatus in the polling area during the 48 hours ending with the close of poll; ECI applies it across media; platforms process s.126 violation reports within 3 hours under the 2019 Voluntary Code of Ethics	**Silence-period calendar guard**: blocks scheduled publishing workflows and flags any paid/boosted items still running; internal analytics continue	L9, L10
Exit polls	RP Act s.126A: no conducting/publishing exit polls during the ECI-notified window (from start of first phase to 30 minutes after close of last phase); s.126(1)(b) also covers opinion-poll display in the 48-hour window	Platform does **not** conduct exit polls; survey outputs are internal; restricted report types disabled in notified windows	L9
Expenditure	Candidate expenditure limits (notified Jan 2022): ₹95 lakh (PC, larger states) / ₹75 lakh (smaller states); ₹40 lakh (AC, larger states) / ₹28 lakh (smaller states) **[VERIFY current limits]**; digital spend must be reported	Spend tracker by category with configurable limits and threshold alerts; reconciliation with finance	L11
Platform policies	Meta requires authorisation + "Paid for by" disclaimers for ads about social issues, elections or politics; Google requires India election-ads verification and an ECI/authorised pre-certificate per ad	Platform-policy checklist per ad; transparency-library reconciliation	L16, L17
Data protection	DPDP Act 2023 + DPDP Rules 2025 (phased)	§9 governance; consent tracking; breach runbook	L12, L13
Cyber incident reporting	CERT-In Directions (28 Apr 2022): report specified incidents within 6 hours; maintain ICT logs 180 days in India	Security runbook; log residency	L14
Misinformation / disinformation	ECI works with State IT Nodal Officers and platforms; April 2026 reporting cited >11,000 violative posts actioned since 15 Mar 2026 **[VERIFY]**	Misinformation-*risk* flags → human verification → escalation to legal/comms; platform never auto-declares content "fake"	L6

10.2 Election Compliance Control Tower

Purpose: give the Compliance Officer and Agency leadership one auditable view of every regulated item — and make non-compliance visible *before* it happens.

Tracked object model: `content_item → approval → certification → ai_provenance → publication → spend → takedown_event`, each with owner, timestamp and immutable audit entries.

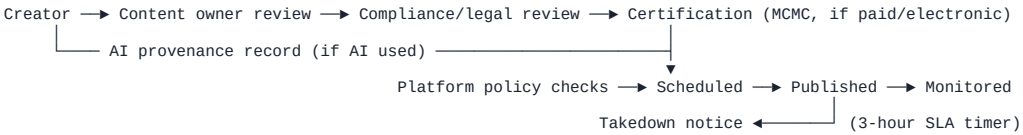
Field	Description	Example (DEMO)
Content ID	Unique ID for creative/ad/post	CT-0417
Type	Organic post / paid ad / bulk SMS / print / AV	Paid ad — video
Content approval status	Draft → Legal review → Approved / Rejected	Approved
Approval owner	Named approver	Compliance Officer (Agency)
Approval timestamp	IST	2026-09-12 16:42
Advertisement certification status	Not required / Applied / Certified / Rejected; MCMC reference	Certified — MCMC-DEMO-0417
Platform policy status	Disclaimer/authorisation done per platform	Meta ✓ / Google ✓
Spend tracking	Committed / delivered spend; category	₹1.8 lakh committed
AI-content disclosure/label status	Not AI / AI-assisted (non-deceptive edit) / AI-generated — label applied ✓/✗; label text; responsible-entity disclosure	AI-generated — label ✓ "Synthetic Content", entity in caption ✓

Field	Description	Example (DEMO)
AI provenance	Tool, creator, timestamp, source assets, consent for likeness	Tool X; creator ID; consent ref
Compliance alerts	Missing certification, label missing, silence-period conflict, spend threshold	None
Takedown events	Notice time, action time, SLA (3 h)	—
Audit trail	Every state change, who/when/why	7 events

Rules engine (examples, configurable by legal):

- `PAID_ITEM_WITHOUT_CERT_REF` → **Critical**: block “ready to publish” status.
- `AI_FLAG = true AND label_applied = false` → **Critical**.
- `scheduled_publish_time IN silence_window(constituency)` → **Critical**: block and notify.
- `spend_to_date / configured_limit > 0.8` → **High**; > 0.95 → **Critical**.
- `takedown_notice_received AND elapsed > 2h AND not actioned` → **Critical** (before 3-hour limit).
- `approval_age > 7 days AND not published` → **Low** (re-review stale approvals).

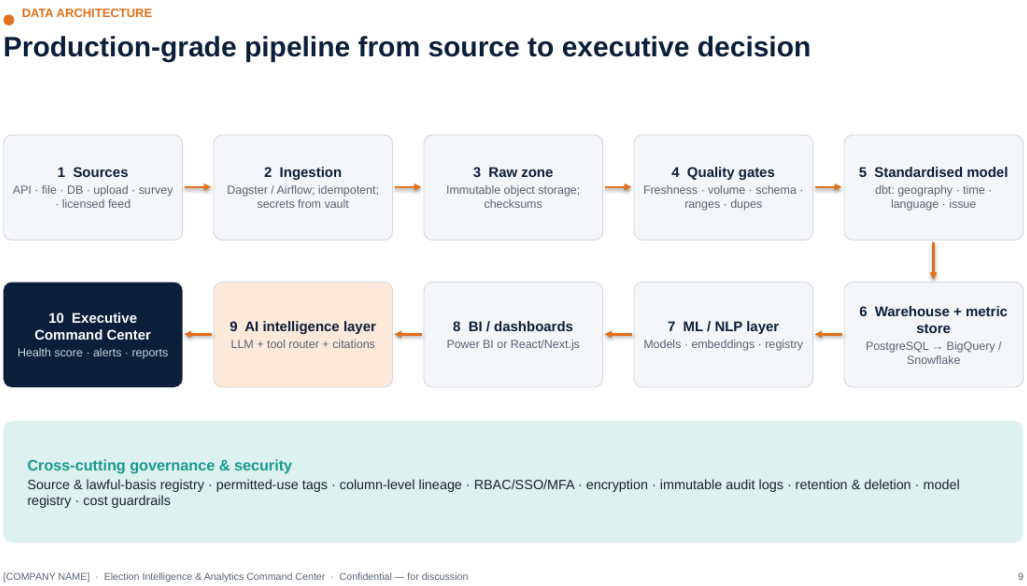
Approval workflow:



The tower **supports** the compliance team; it does not make legal determinations and does not replace the Campaign’s statutory filings.

11. Data Engineering Architecture

11.1 Reference architecture



Data architecture (from the executive deck)

```
SOURCES            Public/official · Campaign-owned · Survey · Digital platforms · Media (licensed) · Field · Call-centre
  |
ACCESS MODES       API connector (verified) · file drop · DB replication · validated upload · survey export · licensed feed
  |
INGESTION          Orchestrated jobs (Dagster/Airflow) · idempotent loads · schema contracts · secrets from vault
  |
RAW ZONE           Object storage, immutable, partitioned by source/date, checksum + ingestion metadata
  |
DATA-QUALITY       Tests: freshness · volume · schema · nulls · ranges · duplicates · integrity → quarantine on fail
  |
STANDARDISED MODEL dbt models: staging → conformed dimensions (geography, time, language, issue, channel) → facts
  |
WAREHOUSE          PostgreSQL (POC / single AC) → BigQuery or Snowflake (multi-AC / state) · metric store
  |
ML / NLP LAYER     Batch & streaming inference · model registry · feature tables · embeddings (pgvector / FAISS)
  |
BI / DASHBOARD     Power BI (fast rollout) or React/Next.js (bespoke Command Center) · Streamlit for POC prototypes
  |
AI INTELLIGENCE    LLM + tool router (metrics API, SQL templates, search, vector retrieval) · guardrails · citation builder
  |
EXECUTIVE COMMAND CENTER   Health score · 14 modules · alerts · reports · AI Analyst · compliance tower
```

11.2 Technology choices — and why each exists

We deliberately avoid “every tool”. Each component earns its place:

Layer	Recommended	Why it exists	When to choose alternatives
Language	**Python + SQL**	Python for ingestion, statistics, ML/NLP; SQL for transformation and analytics — the common language of analysts and engineers	—
Operational DB + POC warehouse	**PostgreSQL** (managed, Indian region)	Reliable, cheap, strong SQL, row-level security, **pgvector** for embeddings — one engine for POC	Move analytics to BigQuery/Snowflake when data > ~1 TB or > 50 concurrent analysts
Scale warehouse	**BigQuery** (GCP) *or* **Snowflake**	Serverless scale, separation of storage/compute, strong governance	Choose per the Agency’s cloud preference and existing contracts
Cloud	**AWS / Azure / GCP** — Indian regions	Data residency, managed security services	Follow client preference; avoid multi-cloud in POC
Orchestration	**Dagster** (asset-centric, built-in lineage) *or* **Airflow** (widely known)	Scheduling, retries, dependencies, observability	Airflow if the client’s team already runs it
Transformati	**dbt**	Versioned SQL models, tests,	—

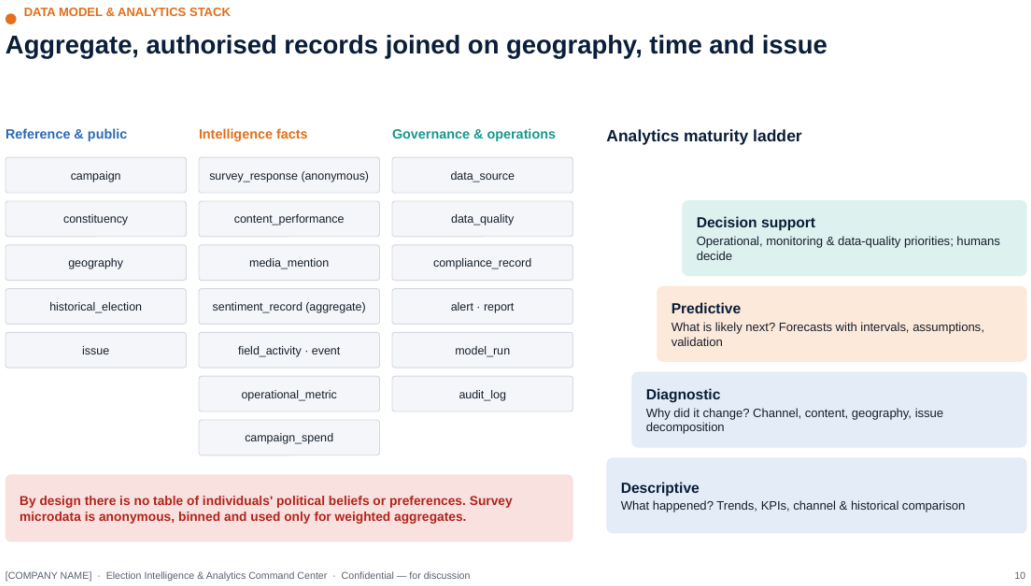
Layer	Recommended	Why it exists	When to choose alternatives
on		documentation, lineage	
Data quality	dbt tests + Great Expectations-style checks	Automated quality gates	—
Statistics	**Pandas, NumPy, SciPy, Statsmodels**	Weighting, CIs, significance tests, time-series	—
ML	**scikit-learn**, gradient boosting (LightGBM/**XGBoost** where it beats simpler models)	Tabular models with explainability	PyTorch only where deep models are justified
NLP	**Transformers**, multilingual encoders (e.g., MuRIL, IndicBERT, XLM-R-class), **IndicTrans2** for translation	Indian-language coverage incl. Hinglish/romanised text	Commercial APIs where accuracy is proven on our gold sets and data terms allow
Embeddings / retrieval	**SentenceTransformers** (multilingual models) + **pgvector** (POC) / **FAISS** (large, in-memory)	Semantic search, clustering, duplicates, RAG	Managed vector DB only at very large scale
API	**FastAPI**	Typed, fast, OpenAPI docs, easy auth integration	—
BI	**Power BI** (Agency-friendly, RLS, mobile) *or* **React/Next.js**	Leadership dashboards	React for a bespoke, branded Command Center; Power BI for speed
Prototype UI	**Streamlit**	1–2 day iterations during POC	Not for production leadership use
LLM	Enterprise LLM endpoint with no-training-on-data terms and Indian/approved region where available	AI Analyst & summarisation	Self-hosted open model where data terms require
Observability	Cloud monitoring + OpenTelemetry; pipeline dashboards	Know when data or models break	—
Secrets	Cloud KMS + secrets manager / Vault	No credentials in code	—

11.3 Metric store (semantic layer) — *added capability*

Every KPI (e.g., `engagement_rate`, `answer_rate`, `activity_completion_rate`) is defined **once** — formula, grain, filters, owner, version — and consumed identically by dashboards, reports and the AI Analyst. This single addition removes the most common source of leadership distrust: “why does your number differ from theirs?”

12. Conceptual Data Model

Design rule: the model stores **aggregate or appropriately authorised operational records**. There is **no table of individuals’ political beliefs, preferences or vote intentions**. Survey microdata is anonymous/pseudonymised, holds coarse demographic bins only, and is used for weighted aggregate estimates.



Conceptual ERD (from the executive deck)

```
campaign ──┬──< constituency >── geography (state/district/AC/block/zone; official codes)
           │          └──< historical_election (year, turnout, electors, party vote share — public)
           ├──< data_source >──< data_quality (test, status, score, run_at)
           ├──< survey >──< survey_question
           │          └──< survey_response (anonymous, weight, bins, QC flags) ──> issue
           ├──< digital_channel >──< content >──< content_performance (daily)
           │                              └──< compliance_record (approval, MCMC ref, AI label, audit)
           ├──< campaign_spend (channel, category, amount, certification ref)
           ├──< media_mention (outlet type, language, topic, tone) ──> issue
           ├──< sentiment_record (aggregated social: date × platform × topic × language × zone*) ──> issue
           ├──< field_activity (zone, type, planned, completed, attendance est.) ──> event
           ├──< operational_metric (call-centre & ops aggregates)
           ├──< alert (rule, severity, status, owner, ack/closure)
           ├──< report (type, period, version, approver, distribution)
           └──< model_run (model, version, data window, metrics, approval)
  * zone only where public location metadata exists and is permitted
```

Entity	Grain	Key fields	Sensitivity
campaign	One per engagement	name, phase, dates	Low
constituency	AC/PC	official code, name, state, reserved category (public)	Public
geography	Admin unit	code, parent, level, geometry ref, boundary source/version	Public
historical_election	AC × election	electors, votes polled, turnout %, party vote share (public)	Public
survey	Wave	design, mode, fieldwork dates, vendor, target n, weighting scheme	Internal
survey_question	Question	wording (all languages), scale, version	Internal
survey_response	Anonymous respondent × wave	age band, gender, area type, zone, answers, weight, QC flags, consent flag	Confidential (pseudonymised)
digital_channel	Account/page/channel	platform, owner, authorised token ref	Internal
content	Creative/post	format, theme, language, length, publish time, AI flag	Internal
content_performance	Content × day	impressions, reach, engagement, views, watch time, clicks, spend	Internal
media_mention	Article/segment	outlet, type, language, topic, tone, reach tier, URL, licence ref	Internal
issue	Taxonomy node	name, parent, keywords/examples, languages	Internal
sentiment_record	Date × platform × topic × language (× zone)	counts by label, mean score, coverage	Internal (aggregate)
field_activity	Activity	zone, type, planned/completed, attendance estimate, report timeliness	Internal
event	Event	type, zone/site, date, estimated attendance	Internal
operational_m	Metric × day × unit	call volume, answer rate, FCR, backlog	Internal

Entity	Grain	Key fields	Sensitivity
etric			
campaign_spend	Line item	channel, vendor, category, amount, certification ref	Confidential
compliance_record	Content × control	approval, certification, AI label, takedown, audit	Confidential
data_source	Source	registry fields (§9)	Internal
data_quality	Test run	test, status, rows affected, score	Internal
alert	Alert	rule, severity, entity, status, owner, timestamps	Internal
report	Report	type, period, version, approver	Internal
model_run	Run	model, version, window, metrics, approval	Internal

A conceptual DDL is provided in `poc/schema.sql`.

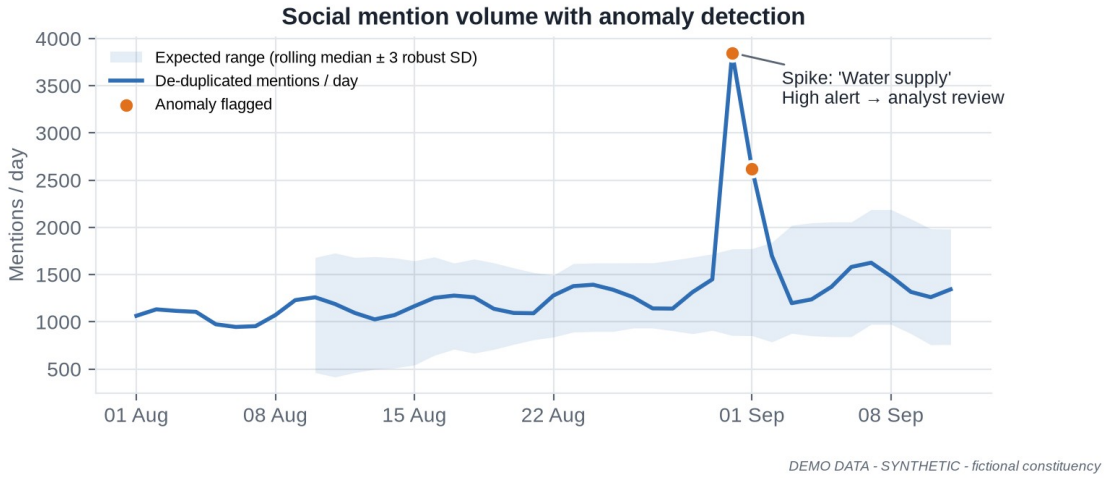
13. Analytics Layer & Intelligence Modules

13.1 Analytics catalogue

Level	Question	Examples	Methods	Output guardrail
Descriptive	What happened?	Daily/weekly/monthly trends; KPI summaries; channel performance; historical comparison vs previous elections (turnout, electors)	Aggregations, rolling baselines, indexed comparisons	Show “data as of”, source, quality
Diagnostic	Why did a KPI change? Which factor/content/channel/geography changed?	Engagement down 18% W/W → driven by X platform, static images, 2 themes; completion drop concentrated in Hill Block	Contribution/variance decomposition, drill-down, mix-vs-rate analysis, change-point detection	Label as “contributing factors”, not causes, unless tested
Predictive (where statistically justified)	What is likely next?	Issue-volume and media-volume forecasts; digital performance forecasts; turnout-*trend* estimates at AC level from historical/public data; campaign operations forecasts (completion, backlog)	Time-series models, regression, gradient boosting	Every forecast shows interval, assumptions, training window, validation metric, limitations; **never certainty; never individual-level**
Decision support	What should we look at / prioritise operationally?	Operational priorities (zones behind plan), data-quality priorities, monitoring priorities (rising issues), resource-measurement insights, content-performance learnings, recommended analytical follow-ups	Rules + ranking + analyst judgement	**No individualised persuasion instructions**; recommendations are operational/analytical, owned by humans

13.2 Social Listening module

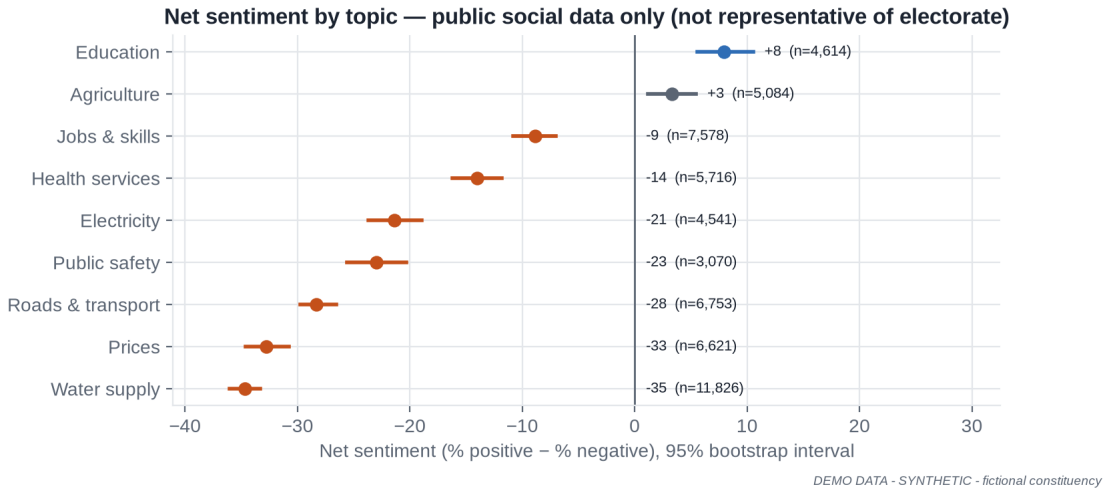
Pipeline: Social Listening → NLP → Topic → Sentiment → Trend → Alert.



Social listening — anomaly detection on DEMO DATA

Dashboard component	Definition	Warning shown
Mention volume	De-duplicated public mentions matching the campaign's tracked keywords/entities	Coverage index; platform mix
Sentiment	Distribution of positive/neutral/negative with model confidence; net sentiment with bootstrap CI	"Public social data only — not representative of the electorate"
Topic trends	Share of mentions by issue taxonomy over time	Classifier accuracy per language
Issue velocity	Growth rate of an issue vs its 7-day baseline (z-score / robust MAD)	Minimum-volume threshold
Engagement	Interactions on public posts in scope	Platform-reported; not comparable across platforms without indexing
Content performance	Top public posts by engagement (aggregate)	No profiling of individual posters
Creator/media amplification	Accounts/outlets driving *reach* on a topic — public figures, media houses, verified organisations	**Aggregated by account type**; no private individuals profiled
Platform comparison	Volume/sentiment split by platform	Access differs per platform
Geographic aggregation	Zone/district only where public location metadata exists and use is permitted	Typically < 20% of posts carry usable location — shown explicitly
Time-series movement	Hourly/daily lines with anomaly bands	—
Emerging-issue detection	New clusters of semantically similar posts growing faster than baseline	Human analyst validates before escalation

Coverage & Representativeness Index (added capability) — each tile shows: platforms covered, % of posts with language confidence ≥ threshold, % geo-attributable, estimated duplicate share, and a standard caveat. In the DEMO data: 18.1% geo-attributable, 7.0% near-duplicates removed.

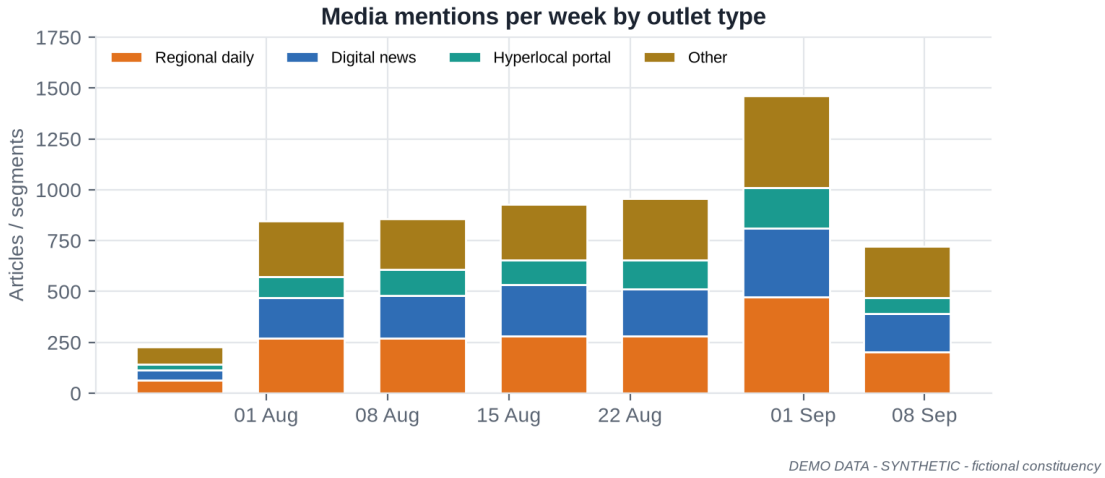


Net sentiment by topic with 95% intervals — DEMO DATA

13.3 Media Intelligence module

Component	Description
Media monitoring	Ingest licensed/public news feeds, regional outlets, broadcast transcripts (licensed)
Article classification	Issue taxonomy; article type (news, opinion, interview, explainer)
Topic extraction	Keyphrases, clusters, story threads across outlets
Sentiment / tone	Tone towards tracked entities/issues, with confidence
Candidate/organisation mention	NER + entity linking to a curated list of public figures and organisations
Issue tracking	Share of voice by issue, outlet type, language, region
Media-volume trend	Daily/weekly volume with anomaly detection
Regional media monitoring	Indian-language outlets prioritised; translation for leadership summaries
Broadcast transcript analysis	**Only where licensed**; transcript segments classified like articles
Alerting	Spike in volume/negative tone for an issue; high-reach outlet coverage

Media Intelligence Dashboard layout: KPI strip (articles 24h / 7d, % negative, share of regional coverage, top issue) → weekly volume by outlet type → issue × tone matrix → story-thread list with outlet count and first-seen time → “coverage map” by district (where outlet geography is known) → daily AI summary with citations.



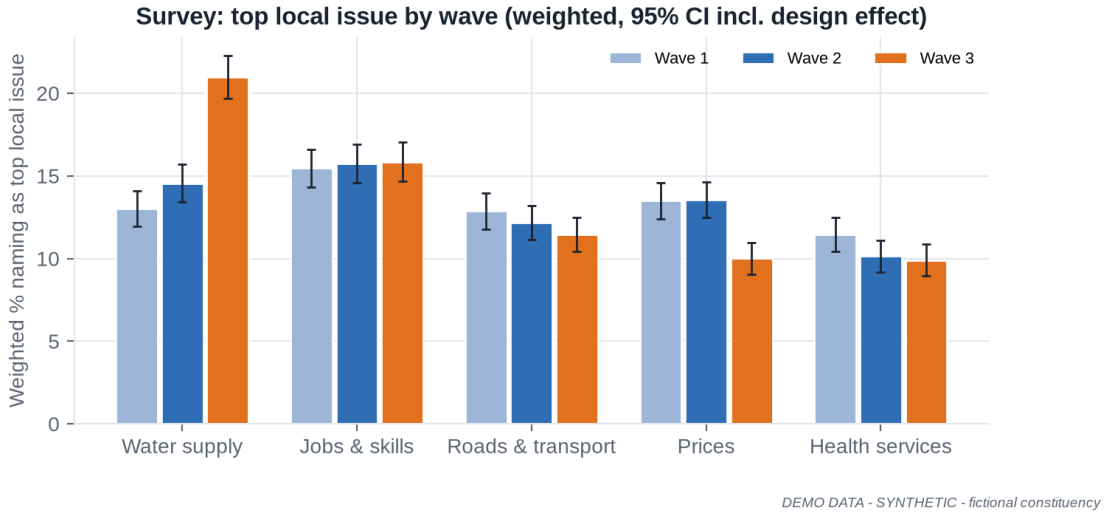
Weekly media volume by outlet type — DEMO DATA

13.4 Survey Intelligence module

Capability	Specification
Questionnaire design support	Neutral-wording checklist, scale balance, order rotation, translation/back-translation, cognitive

Capability	Specification
	pre-test log, sensitive-question review (no vote-intention questions unless the Campaign's research lead and legal approve and outputs remain aggregate and internal)
Sampling	Design registry (stratified/cluster/quota), sampling frame description, allocation by stratum
Sample-size calculation	$n = z^2 \cdot p(1-p)/e^2 \times deff$, with finite-population correction where relevant; calculator in the UI
Weighting	Post-stratification / raking to Census/official margins; trimming (e.g., 0.3–3.0); weight diagnostics
Confidence intervals & MoE	Using effective sample size; shown by default
Cross-tabulation	Weighted cross-tabs with column %, n and MoE; suppression of cells with n < 50
Statistical significance	Two-proportion z-tests with design effect; wave-over-wave change flags; multiple-comparison control (Benjamini–Hochberg) for large cross-tab batteries
Response quality	Speeders, straight-liners, duplicates, interviewer effects, GPS/time anomalies (tablet F2F), back-check results
Missing data	Item non-response rates; explicit "Don't know/Refused"; imputation only when documented
Bias checks	Sample vs population profile; mode effects; non-response analysis
Trend comparison	Same-question, same-population comparisons only; methodology changes flagged

DEMO example: Wave 3 (n = 3,851; effective n = 3,704 after weighting) shows 21.0% ± 1.3 naming *water supply* as the top local issue vs 13.0% in Wave 1 — a change of +8.0 points, statistically significant (z ≈ 9). Awareness of the campaign's public programme rose from 30.2% to 44.1%. **Surveys describe the sampled population at fieldwork time; they do not guarantee or predict election results.**

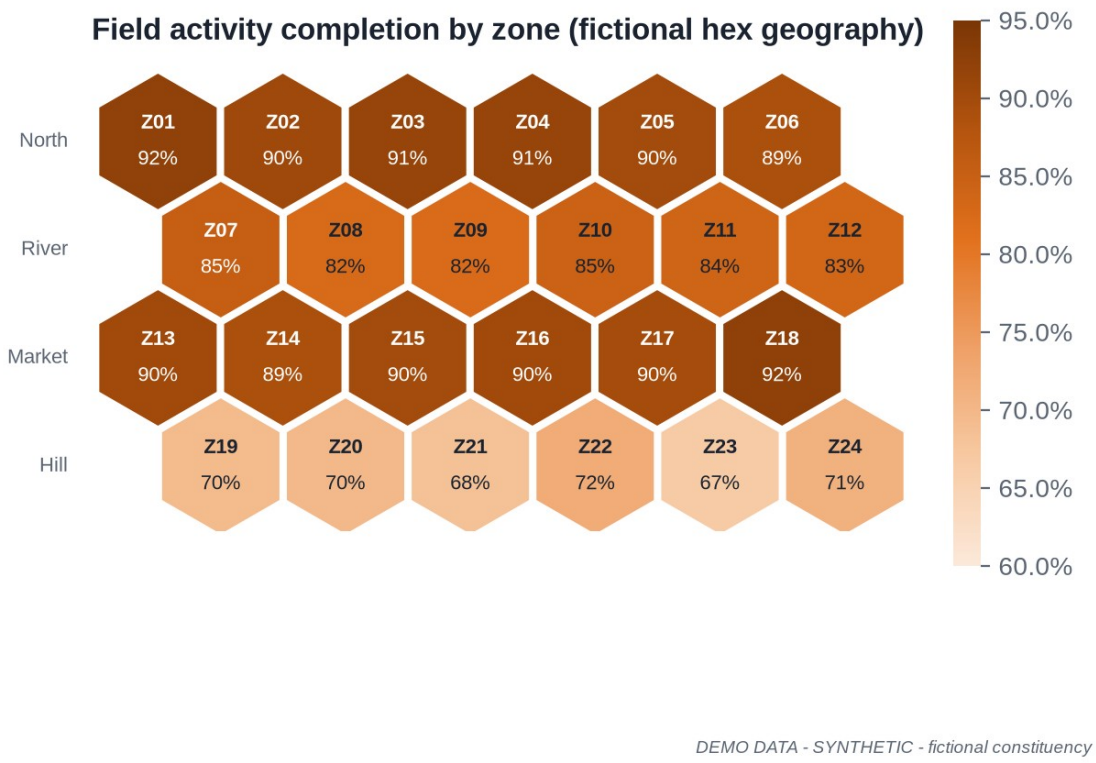


Survey issue salience with 95% CIs — DEMO DATA

13.5 Field Operations Analytics module

KPI	Definition
Activity completion rate	Completed / planned activities
Event count	Events held by type
Attendance (where lawfully recorded)	Headcount *estimates* per event; no identity capture
Geographic coverage	% of zones with ≥ 1 completed activity in last 7 days
Issue-report volume	Local issue reports logged by category
Volunteer activity	Active volunteers, tasks per volunteer (aggregate)
Task completion	On-time task completion %
Response time	Time from issue report to acknowledgement
Data submission quality	Timeliness + completeness + validation pass rate
Operational bottlenecks	Zones/blocks with persistent under-completion or late reports

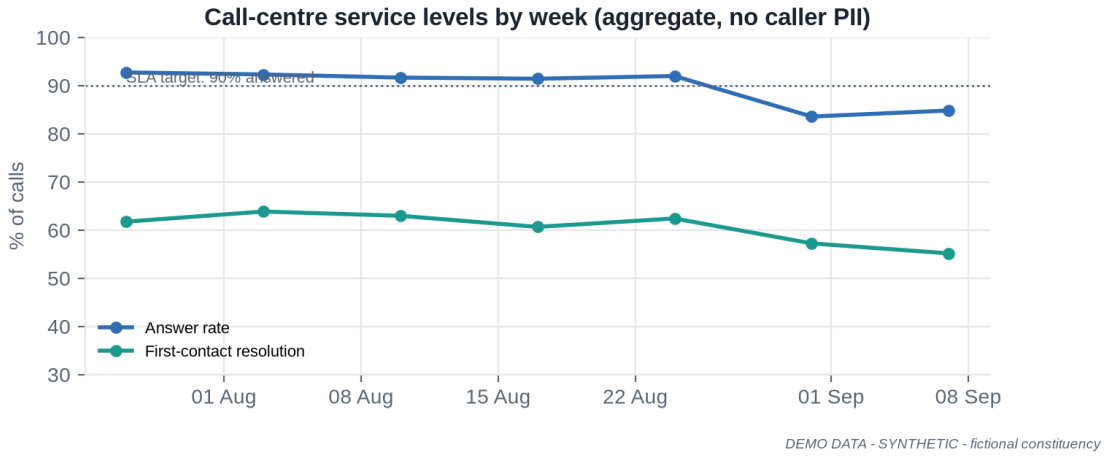
Field Operations Dashboard: completion by zone (map), completion trend, on-time reporting by block, bottleneck list with owner, issue-report categories.



Field completion by zone — DEMO DATA, fictional geography

13.6 Call-centre Analytics module (if authorised data is available)

Metrics: call volume, answer rate, response/wait time, issue categories, resolution rate (first-contact and overall), follow-up status, language mix, geographic aggregation (zone/district), agent performance (operational metrics for the helpline's own management, not surveillance beyond lawful workplace norms), quality-monitoring scores. **Not used:** private communications, unauthorised personal data, or any inference about callers' political views. Recordings/transcripts only where callers are informed and the Campaign's legal team has approved.



Call-centre service levels — DEMO DATA

14. Machine Learning & Multilingual NLP

14.1 Model catalogue

Hard rule: no model infers an individual's political ideology, preference or likely vote. Models operate on aggregates, content, text of public/licensed items, or operational records.

(a) Purpose, inputs, outputs and method

#	Model	Purpose	Input	Output	Method
M1	KPI driver regression	Explain which factors are associated with engagement changes	Content features (format, theme, length, language, hour, platform, boost)	Coefficients / partial effects with CIs	OLS / regularised regression on log engagement
M2	Content "high-performer" classifier	Flag content likely to exceed median ER *for format planning*	Content metadata (no audience attributes)	Probability + drivers	Logistic regression → gradient boosting if lift justified
M3	Issue/topic classifier	Map text to the campaign issue taxonomy	Public/licensed text, survey open-ends, field issue notes	Issue label(s) + confidence	Fine-tuned multilingual encoder; zero-shot fallback
M4	Sentiment/ tone classifier	Aggregate sentiment/tone by topic	Public/licensed text	Pos/neu/neg + confidence	Fine-tuned multilingual encoder
M5	Clustering / emerging-topic detection	Find new story clusters	Text embeddings	Clusters, keyphrases, growth	Embeddings + HDBSCAN/BERTopic-style topic modelling
M6	Near-duplicate detector	De-duplicate; spot copy-paste amplification	Text, embeddings, MinHash	Duplicate groups	MinHash/LSH + cosine similarity
M7	Time-series forecaster	Issue-volume, media-volume, digital KPI, ops forecasts	Daily series + calendar features	Point forecast + 80/95% intervals	Seasonal regression / ETS / Prophet-class; gradient boosting for multi-series
M8	Anomaly detector	Spikes/drops in any KPI	Time series	Anomaly score, severity	Robust rolling median ± k·MAD; STL residuals; isolation forest for multivariate
M9	Turnout-trend model (aggregate)	Contextualise AC-level turnout trends for **operational planning**	Public historical turnout, electors, polling stations, public development indicators	Expected turnout range per AC (not per person)	Hierarchical/regularised regression; Random Forest benchmark
M10	Survey weighting & small-area estimates	Robust aggregate estimates	Survey microdata + population margins	Weighted estimates, MoE	Raking; MRP only with strong justification
M11	PCA / factor analysis	Reduce many indicators to interpretable dimensions (e.g., development index)	Public indicators at AC/district	Components, loadings	PCA
M12	NER & entity linking	Identify public figures, organisations, places, schemes	Public/licensed text	Entities + IDs	Multilingual NER + curated gazetteer
M13	Toxicity / safety classifier	Protect analysts; flag abusive content in monitored spaces	Public text	Toxicity score	Multilingual classifier
M14	Misinformation-risk flagger	Prioritise items for human fact-checking	Public text + propagation features	Risk score + reasons	Claim-matching against verified fact-check library + heuristics

(b) Evaluation, risks, limitations and human review

#	Model	Evaluation metric	Key risk	Limitations	Human review
M1	KPI driver	Adj. R^2, out-of-sample	Confounding read as	Associational, not	Analyst signs off

#	Model	Evaluation metric	Key risk	Limitations	Human review
	regression	RMSE, coefficient stability	causation	causal	interpretation
M2	Content “high-performer” classifier	ROC-AUC, calibration (Brier), precision@k	Self-fulfilling feedback	Platform algorithm changes shift behaviour	Digital lead reviews
M3	Issue/topic classifier	Macro-F1 per language on gold set	Taxonomy drift, code-mixing errors	Weak on rare dialects	Low-confidence items to analyst queue
M4	Sentiment/tone classifier	Macro-F1, per-class recall, per-language F1	Sarcasm, political idiom, bias across languages	Social ≠ population	Monthly re-validation
M5	Clustering / emerging-topic detection	Topic coherence, analyst usefulness rating	Spurious clusters	Needs minimum volume	Analyst validates new topic before alert ≥ High
M6	Near-duplicate detector	Precision/recall on labelled pairs	Over-merging similar genuine posts	Paraphrase-heavy campaigns	Coordinated-copy flags reviewed by analyst
M7	Time-series forecaster	MAPE/sMAPE, interval coverage on hold-out	Event shocks	Short histories	Forecast note signed by DS
M8	Anomaly detector	Precision/recall on labelled incidents; alert fatigue rate	False positives	Low-volume series noisy	Triage by war-room analyst
M9	Turnout-trend model (aggregate)	MAE on held-out past elections, interval coverage	Over-interpretation	Structural breaks (delimitation, roll revisions)	Published only internally with caveats
M10	Survey weighting & small-area estimates	Benchmark comparison, design effect	Model-driven small-area noise	Needs adequate sample per cell	Research lead approves
M11	PCA / factor analysis	Variance explained, loading stability	Misleading labels	Linear	Analyst names components
M12	NER & entity linking	F1 on gold set	Ambiguous names	Transliteration variants	Gazetteer curated by analysts
M13	Toxicity / safety classifier	Precision/recall on gold set	Dialect bias	Context-dependence	Human decision on any action
M14	Misinformation-risk flagger	Precision@k of analyst-confirmed items	Labelling legitimate speech as “fake”	Cannot determine truth	**Always human-verified**; never auto-labelled

14.2 Multilingual NLP layer

Capability	Approach	Notes
Language detection	fastText-class LID + script detection + Hinglish/romanised heuristics	Code-mixed text is the norm, not the exception
Hindi, English, Hinglish	Multilingual encoders (MuRIL, IndicBERT, XLM-R-class) fine-tuned on campaign gold sets	Separate metrics for each
Regional languages	Per-language evaluation; IndicTrans2 (22 scheduled languages, MIT licence) for translation where needed	Prioritise the languages of the target geography
Romanised Indian languages	Transliteration normalisation (e.g., IndicXlit-class) before classification	Major error source if ignored
Spell correction / normalisation	Lexicon + transliteration-aware normalisation	Preserve original text for audit
Topic classification	M3	—
Sentiment	M4	—
NER	M12	—
Event detection	Burst detection on entity + location + keyphrase co-occurrence	Analyst validation
Issue extraction	Taxonomy classification + keyphrase extraction	—
Trend detection	Velocity vs baseline, change-points	—
Duplicate detection	M6	—
Toxicity/safety	M13	—
Misinformation-risk	M14	—
Content classification	Format/theme tagging of owned content	Feeds content matrix

Human-labelled gold sets (added capability). For each priority language (incl. Hinglish and romanised forms) we build a stratified gold set (target ≥ 1,500 items per language for sentiment/topic in production; ≥ 500 in POC), double-annotated with inter-annotator agreement (Cohen's κ ≥ 0.70 target). Models are accepted only if they meet per-language thresholds (e.g., macro-F1 ≥ 0.70 for sentiment in POC; ≥ 0.75 in production) and **results are reported per language**, not just overall.

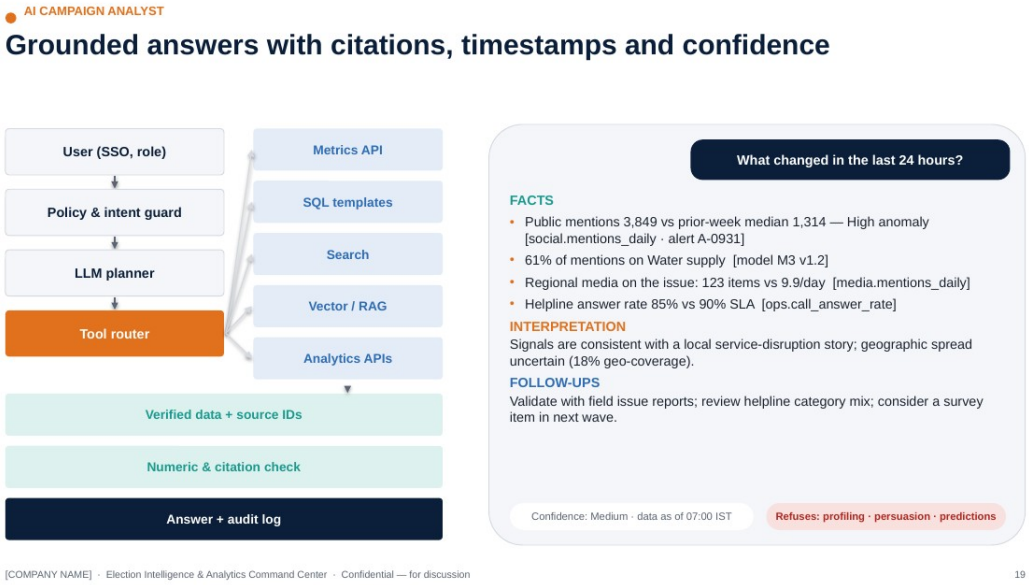
15. AI Campaign Analyst & RAG

15.1 Purpose

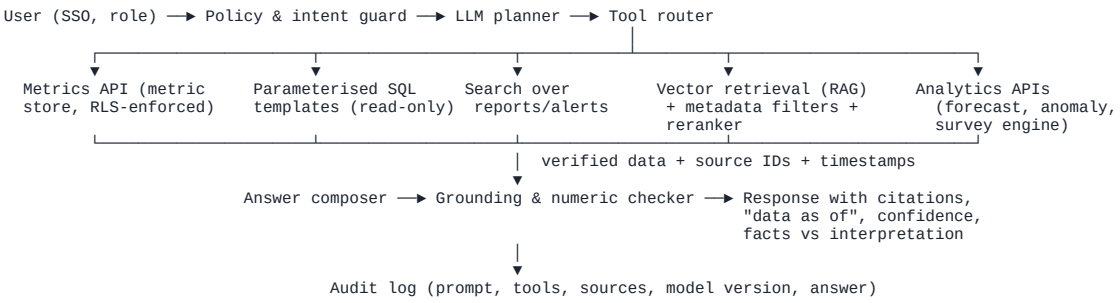
A secure, internal assistant that answers leadership and analyst questions **from approved, governed data only**, e.g.:

- "What changed in the last 24 hours?"
- "Which issues are trending?"
- "Show digital performance versus last week."
- "What data sources have quality issues?"
- "Summarise today's media coverage."
- "Which campaign activities had the highest measured engagement?"
- "Compare this week with the previous four weeks."

15.2 Architecture



AI Analyst architecture (from the executive deck)



15.3 Behavioural contract

The AI Analyst **must**:

1. Retrieve source data through tools — never answer numeric questions from model memory.
2. Cite internal sources (dataset/metric ID, report ID, document + page) for every factual claim.
3. Show the data timestamp ("data as of 07:00 IST").
4. Show confidence (High/Medium/Low) based on data quality, coverage and model confidence.
5. Separate **Facts** (from data) and **Interpretation** (analyst-style reading), labelled.
6. Never fabricate numbers; the numeric checker verifies every number in the answer appears in tool outputs (or is a deterministic calculation from them).
7. Never invent sources; citations are generated from tool metadata, not by the LLM.
8. Never make unsupported political predictions (e.g., seat or vote-share predictions) and never produce individual-level profiling or persuasion scripts — it declines and explains.
9. Respect RBAC: the user sees only what their role permits; retrieval is filtered by permissions.

Example (DEMO):

> **Q:** What changed in the last 24 hours?
>
> **Facts** — (1) De-duplicated public mentions: 3,849 on 31 Aug vs prior 7-day median 1,314 — flagged **High** anomaly *[metric: social.mentions_daily; alert A-0931; data as of 01 Sep 07:00]*. (2) 61% of that day's mentions relate to *Water supply [model M3 v1.2]*. (3) Regional/hyperlocal media items on the same issue: 123 in 24h vs a prior daily average of 9.9 *[media.mentions_daily]*. (4) Helpline answer rate on 31 Aug: 85% (SLA 90%) *[ops.call_answer_rate]*.
>
> **Interpretation** — The social and media signals are consistent with a local service-disruption story. Social data covers public posts only (18% geo-attributable), so geographic spread is uncertain.
>
> **Suggested analytical follow-ups** — Validate with field issue reports from affected zones; check call-centre category mix; add a question in the next survey wave if the issue persists.
>
> *Confidence: Medium (coverage limits). Sources: 4 datasets, 1 alert.*

15.4 RAG knowledge base

Collection	Contents (authorised only)	Metadata	Access
Campaign reports	Daily/weekly reports, analyses	date, type, author, approval	Role-based
Historical reports	Past campaign reviews	year, geography	Role-based
Survey methodology	Questionnaires, sampling, weighting notes	wave, vendor	Research + leadership
Public election documents	ECI statistical reports, handbooks, MCC compendia, press notes	issuing body, date, URL	All
Campaign SOPs	War-room SOPs, escalation matrix	version	All
Content-performance reports	Test results, learnings	period, channel	Digital + leadership
Media reports	Daily summaries	date	Media + leadership
Compliance documents	Approved policies, legal guidance received	version, approver	Compliance + leadership

Pipeline: document intake (approval required) → parsing/OCR → chunking by structure → multilingual embeddings → vector index (pgvector/FAISS) + BM25 hybrid → **metadata filtering** (role, date, doc type) → **reranking** (cross-encoder) → **grounded generation** with quote-level citations → citation verification → answer. Every answer carries **source traceability** to document ID and page/section.

15.5 Hallucination protection & red-team harness (added capability)

Control	Mechanism
Tool-first answering	Numeric intents must call a tool; no tool → "I don't have verified data for that"
Numeric grounding check	Regex-extract numbers from the draft; each must match tool output (tolerance for rounding)
Citation integrity	Citations inserted from tool metadata; LLM cannot author citation strings
Refusal policy	Classifier + rules for out-of-policy requests (individual profiling, persuasion scripts, outcome

Control	Mechanism
	predictions, legal conclusions)
Evaluation set	200+ questions (POC: 60) with expected sources; measured weekly: answer accuracy, citation validity, refusal correctness
Red-team prompts	Prompt-injection in retrieved documents, requests for PII, “predict the winner”, “write a message to persuade X”, jailbreaks
Versioning	Prompts, tools and model versions logged per answer; regression tests on change

16. Security

Control	Specification
RBAC	Roles mapped to SSO groups; least privilege; row-level security by geography/team; column masking for sensitive fields
SSO	SAML 2.0 / OIDC with the Agency’s identity provider
MFA	Mandatory for all users; phishing-resistant (FIDO2/passkeys) for admins
Encryption at rest	AES-256 with cloud KMS; customer-managed keys in Enterprise tier
Encryption in transit	TLS 1.2+ everywhere; HSTS; mTLS for service-to-service in production
Secrets management	Cloud secrets manager/Vault; no secrets in code, notebooks or chat; automated secret scanning in CI
Audit logs	Centralised, immutable; retention per §9; alerts on anomalous access (bulk export, off-hours admin)
API key rotation	≤ 90 days; immediate on staff exit or suspected exposure
Environment separation	Separate dev/test/prod accounts/projects; prod data never in dev; synthetic data for testing
Backup	Daily snapshots + PITR; encrypted; cross-zone in-region
Disaster recovery	Documented DR plan; RPO/RTO per §7; semi-annual DR test; poll-week warm standby (Enterprise)
Incident response	24×7 on-call in high-intensity phase; CERT-In 6-hour reporting path; DPDP breach workflow; comms templates
Data retention	Per Source Registry; automated purges
Least privilege	Just-in-time elevation; quarterly access reviews; separate break-glass accounts
Endpoint & people	Managed devices for analysts; DLP on exports; security training; background checks for privileged roles
Application security	OWASP ASVS L2 target; dependency scanning; annual third-party penetration test (pre-high-intensity phase)
LLM security	Prompt-injection defences; output filtering; no training on client data (contractual); data-region controls

17. Responsible AI & Model Governance

17.1 Responsible AI principles

Principle	What it means in this platform
Human oversight	Humans decide; AI drafts, flags and explains. Critical alerts and all external-facing outputs require human sign-off
No fabricated information	Tool-grounded answers; numeric and citation checks
No unsupported prediction	Forecasts only where validated; intervals and limitations always shown; no seat/vote predictions from social data
Explainability	Driver analysis, feature importance, model cards; plain-language explanations for leadership
Source traceability	Every number → metric → dataset → source
Data minimisation	Aggregate early; drop unnecessary identifiers
Privacy	DPDP-aligned; pseudonymisation; purpose limitation
Security	§16
Bias monitoring	Per-language, per-region performance tracking of NLP models; survey representativeness checks
Model monitoring	Drift, performance decay, alert precision

Principle	What it means in this platform
Synthetic-content safeguards	Platform does not generate campaign creative; tracks AI provenance and labels; supports 3-hour takedown workflows
Compliance review	Legal/compliance review of policies, prompts that touch compliance, and any new data purpose

Alignment references: MeitY **India AI Governance Guidelines** (5 Nov 2025; seven guiding principles), **NIST AI RMF 1.0**, **ISO/IEC 42001:2023** (AI management systems).

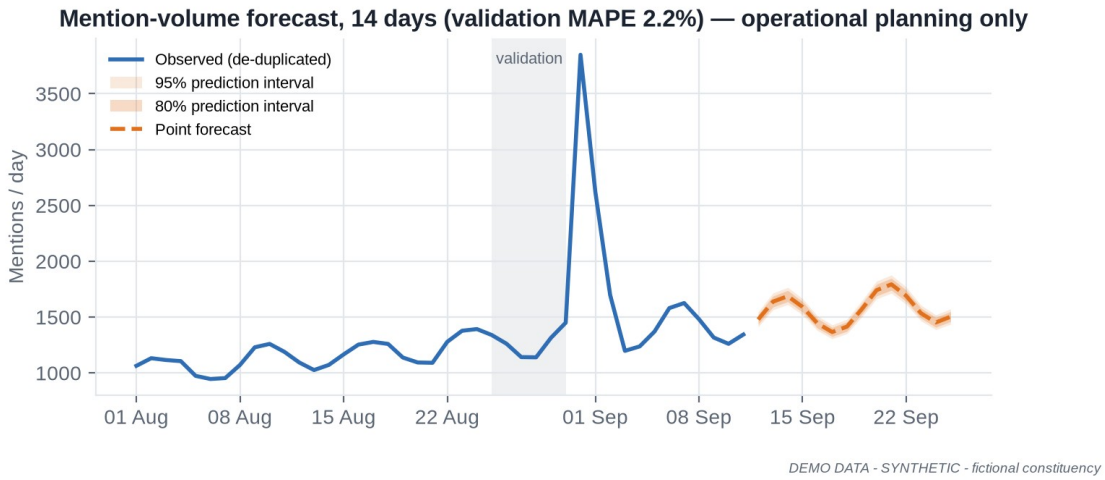
17.2 Three kinds of "AI" — kept strictly separate

Category	Definition	Examples	Governing controls
AI-assisted analytics	AI used internally to analyse data and draft internal insights	Topic classification, forecasts, AI Analyst answers	Model governance, human review, citations. Internal outputs are not campaign content
AI-generated content (general)	Any text/image/audio/video produced or materially altered by AI	Draft summaries, translated reports	Internal-use labelling; human review before any external use
AI-generated campaign creative	Synthetic or AI-altered media published for campaigning	AI-generated video/image/audio ads or posts	**ECI advisory (24 Oct 2025)**: visible label (≥10% area; audio first 10%), responsible-entity disclosure, no deceptive likeness without consent, record-keeping, 3-hour takedown; IT Rules 2026 SGI labelling. **Out of the platform's generation scope** — the Control Tower only tracks provenance, labels and approvals

17.3 Model governance — model card template (required for every model)

Field	Example (M7 — issue-volume forecaster, DEMO)
Model name	issue_volume_forecaster
Purpose	14-day forecast of daily de-duplicated mention volume for monitoring-capacity planning
Dataset	social.mentions_daily (DEMO), anomalies excluded from fit
Training period	01–24 Aug 2026 (validation 25–30 Aug)
Features	Linear trend, weekly Fourier terms
Target	log(daily mentions)
Algorithm	OLS seasonal regression (benchmark: ETS)
Metrics	Validation MAPE 2.2% (synthetic data — real-world error will be higher); 80/95% interval coverage tracked
Validation	Rolling-origin back-test; residual diagnostics
Bias checks	N/A for aggregate volume; platform-mix drift monitored
Limitations	Cannot anticipate event shocks; platform access changes break continuity
Owner	Lead Data Scientist
Version	0.3.0
Last updated	2026-09-12
Approval status	Approved for internal planning use (POC)

Models move through **Draft** → **Validated** → **Approved** → **Monitored** → **Retired**, with approvals recorded in the model registry and every `model_run` logged.

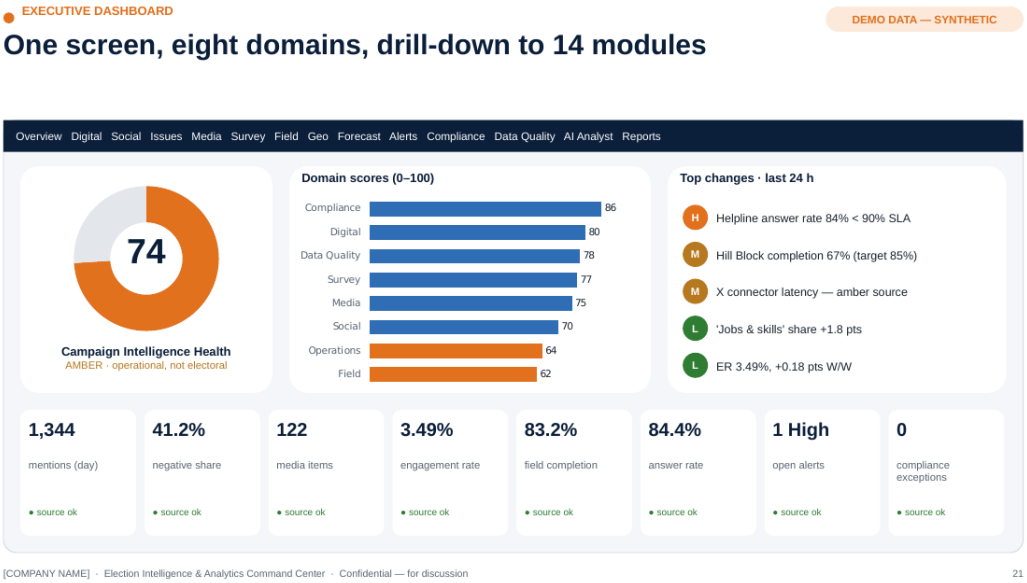


Forecast with prediction intervals — DEMO DATA

18. Command Center UI

18.1 Design principles

- **Answer first, detail on demand.** Every screen opens with 3–5 KPI cards and a one-line “what changed” statement, then charts, then tables.
- **Trust signals on every tile:** source, “data as of”, data-quality badge (green/amber/red), and coverage note where relevant.
- **Uncertainty is visible:** intervals on estimates; significance markers on changes; forecasts shaded.
- **Consistent grammar:** same period selector, geography filter and language filter across modules, in one row above the charts.
- **Accessible:** WCAG 2.1 AA, colour-blind-safe validated palette, status never shown by colour alone.



Executive Command Center mock-up — DEMO DATA

18.2 Home — Campaign Intelligence Health Score

A composite 0–100 index summarising eight Level-2 domains (Digital, Social, Media, Survey, Field, Operations, Data Quality, Compliance). It is an **operational health indicator**, not an electoral prediction.

- Each domain score = weighted average of normalised KPIs vs target/baseline (weights agreed with leadership in Week 1, versioned).
- The Health Score is displayed with its domain breakdown; a red domain cannot be hidden by green ones (any domain < 50 caps the headline at “Amber”).
- Data Quality and Compliance scores gate trust: if Data Quality is red, the score is shown with a “low-confidence” banner.

18.3 Screen-by-screen wireframe descriptions

#	Screen	Layout (top → bottom)	Primary users
1	**Executive Overview**	Header: Health Score dial + 8 domain chips · “Top 5 changes in last 24h” list with severity icons · KPI strip (mentions, net sentiment, media volume, digital ER, field completion, answer rate, open alerts, compliance exceptions) · 14-day trend small multiples · AI daily brief with citations	Leadership
2	**Digital Intelligence**	KPI strip (reach, impressions, ER, video views, avg watch %, CTR, CPC, CPM, spend) · platform comparison bars · Digital Performance Scorecard table vs targets · spend pacing vs plan · top/bottom content	Head of Digital

#	Screen	Layout (top → bottom)	Primary users
3	**Social Listening**	Coverage index banner · mention volume with anomaly band · sentiment distribution with CIs · topic share stream · platform split · emerging-cluster cards (keyphrases, growth, sample public posts)	Social analyst
4	**Issue Intelligence**	Issue taxonomy tree · cross-source issue matrix (social share, media share, survey salience, field reports, call-centre categories) · issue velocity leaderboard · issue detail drawer with timeline across sources	Research, leadership
5	**Media Intelligence**	KPI strip · weekly volume by outlet type · issue × tone heatmap · story threads with first-seen time and outlet count · regional coverage by district · licensed broadcast segments	Media cell
6	**Survey Intelligence**	Wave selector · topline with MoE · wave-over-wave change with significance markers · weighted cross-tabs (with suppression) · sample health (n, n_eff, deff, QC exclusions, profile vs population) · methodology panel	Research lead
7	**Field Operations**	KPI strip (planned, completed %, on-time %, attendance est., issue reports) · zone map · completion trend by block · bottleneck table with owners · data-submission quality	Field head
8	**Geographic Intelligence**	Layer switcher (historical results, turnout trend, survey coverage, activity coverage, events, issue distribution, media distribution, operational coverage) · AC/block/zone choropleth · side panel with selected-area profile (public + operational aggregates)	Leadership, field
9	**Forecasting**	Model selector · forecast chart with 80/95% intervals · model card summary (training window, validation metric, assumptions, limitations) · back-test accuracy history	Analysts
10	**Alerts**	Severity-sorted queue · alert detail (rule, evidence chart, affected entity, owner, SLA timer) · acknowledge / escalate / close with note · alert precision stats	War-room
11	**Compliance**	Exceptions banner · content pipeline Kanban (Draft → Review → Certified → Scheduled → Live) · certification status table · AI-label/provenance ledger · spend vs configured limits · silence-period calendar · audit trail search	Compliance officer
12	**Data Quality**	Source registry health grid (freshness, volume, tests passed) · failing tests · lineage explorer · quality score trend	Data team
13	**AI Analyst**	Chat with suggested questions · answers with Facts / Interpretation / Follow-ups, citations and “data as of” · “show the data” expanders · feedback buttons	All (role-scoped)
14	**Reports**	Daily/weekly report library · draft → review → approve workflow · export (PDF/PPTX) · distribution log	Leadership, account lead

19. Real-Time Alert Engine

19.1 Severity levels

Level	Definition	Response target	Channel
Critical	Compliance breach risk, pipeline outage on priority source, security incident, or extreme anomaly on a leadership KPI	Acknowledge ≤ 15 min; action ≤ 1 h (compliance takedown clock: < 3 h)	Phone/SMS + app + war-room screen
High	Significant anomaly (e.g., > 4 robust SD) on priority metrics; fast-growing issue; high-reach media story	Acknowledge ≤ 1 h	App + messaging integration
Medium	Notable movement (3–4 robust SD) or sustained trend change; data-quality degradation	Same day	App + daily report
Low	Informational; minor deviations; stale approvals	Weekly review	Digest

19.2 Alert catalogue

Alert	Rule (default; tunable)	Default level
Sudden social-volume spike	Daily/hourly mentions > rolling median + 3·(1.4826·MAD) and	High (Critical if > 6 SD and top-3

Alert	Rule (default; tunable)	Default level
	volume > minimum	issue)
Abnormal sentiment movement	Net sentiment change beyond 99% bootstrap interval of 14-day baseline, n ≥ 300	Medium/High
Viral content	Public item in scope with engagement velocity in top 0.1% of 30-day distribution	High
Unusual digital-performance movement	ER/CTR/CPM outside forecast 95% interval for 2 consecutive periods	Medium
Media spike	Media items on an issue > baseline + 3 SD or ≥ 2 high-reach outlets within 6 h	High
Data pipeline failure	Job failed / freshness SLA missed for priority source	Critical (priority) / Medium (other)
Survey anomaly	Interviewer/area outliers; QC exclusion rate > 10%; profile deviation > 5 pts	Medium
Data-quality deterioration	Source quality score drops > 10 pts or below 70	Medium/High
Compliance deadline	Expenditure filing, certification expiry, silence-period start in < 24 h	High/Critical
Abnormal spend	Daily spend > 150% of pacing plan, or cumulative > 80%/95% of configured limit	High/Critical
Unexpected KPI movement	Any metric-store KPI outside its forecast interval	Low/Medium
AI label missing	AI-flagged content without label record	Critical

Alert hygiene: de-duplication and grouping (one incident, many signals), suppression windows, owner assignment, closure notes, weekly precision review (target ≥ 60% of High/Critical alerts rated “useful” by the war-room).

20. KPI Framework

20.1 KPI tree

```
LEVEL 1  Campaign Intelligence Health (0–100, operational)
   |
LEVEL 2  Digital · Social · Media · Survey · Field · Operations · Data Quality · Compliance
   |
LEVEL 3  (detailed, measurable KPIs below)
```

Level 2	Level 3 KPIs	Type
Digital	Reach; impressions; engagement rate; video views; avg watch %; CTR; CPC; CPM; landing-page conversion (sign-ups/volunteer forms — lawful, consented); spend vs pacing	Leading (ER, CTR) / Lagging (reach totals)
Social	De-duplicated mention volume; net sentiment (with CI); issue share; issue velocity; share of voice among tracked entities; emerging-cluster count	Leading / Diagnostic
Media	Media volume; tone distribution; issue share; regional share; high-reach outlet coverage; story-thread count	Leading / Diagnostic
Survey	Issue salience (weighted, MoE); programme awareness; service satisfaction; wave-over-wave significant changes; sample health (n_eff, deff)	Lagging (periodic) / Data-quality
Field	Activity completion %; zone coverage %; on-time reporting %; issue reports logged; response time	Leading (operational)
Operations	Call answer rate; FCR; median wait; backlog; alert acknowledgement time; report on-time delivery	Leading (operational)
Data Quality	Sources green %; freshness SLA %; test pass rate; duplicate rate; coverage index; lineage completeness	Data-quality
Compliance	Paid items with certification ref %; AI items labelled %; approvals within SLA %; silence-period violations (target 0); takedown within 3 h %; spend within configured limits	Leading (risk)

20.2 Indicator types

Type	Purpose	Examples
Leading	Move before outcomes; used for early action	Issue velocity, ER, field completion, alert acknowledgement

Type	Purpose	Examples
Lagging	Confirm outcomes over time	Survey awareness, cumulative reach, weekly satisfaction
Diagnostic	Explain movement	Platform/format/theme mix, zone breakdown
Data-quality	Tell you whether to trust the rest	Freshness, coverage, test pass rate, n_eff

21. Digital Campaign & Content Intelligence

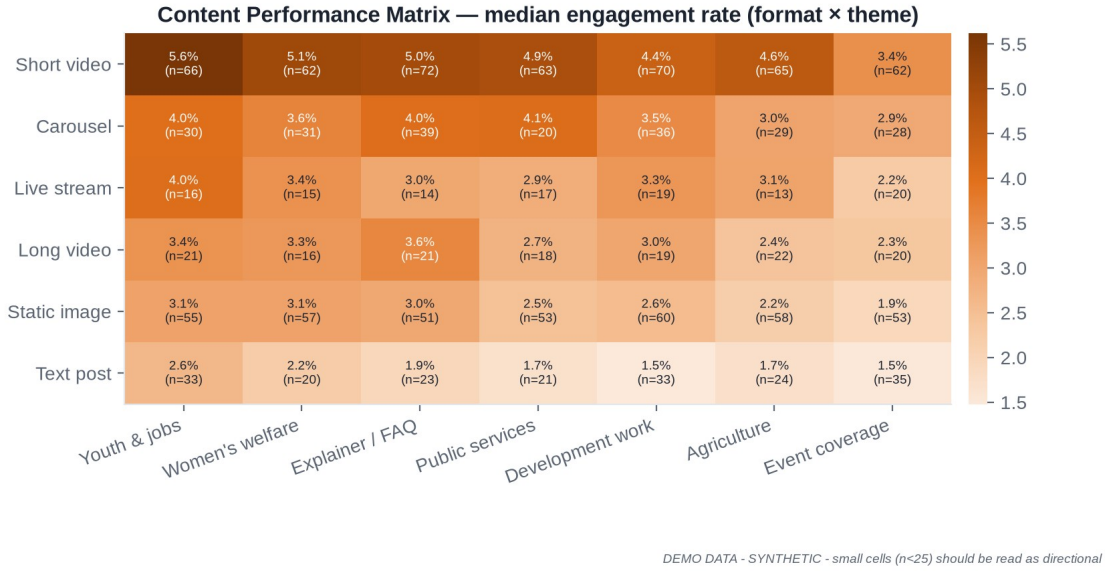
21.1 Digital Performance Scorecard (DEMO DATA, trailing 7 days to 11 Sep 2026 vs prior 7 days)

Metric	This week	Prior week	Δ	Status
Posts published	248	233	+6.4%	—
Impressions	9.21 M	9.13 M	+0.9%	On plan
Engagement rate	3.49%	3.31%	+0.18 pts	▲ Above target (3.2%)
Video views	1.50 M	1.56 M	−3.8%	Watch
CTR	1.11%	1.14%	−0.03 pts	On plan
Paid spend	₹3.04 lakh	₹3.45 lakh	−12.1%	Within pacing
CPM (boosted)	₹116	₹111	+4.5%	Watch
Boosted items with MCMC certification ref	100%	100%	—	✓ Compliant

Definitions are platform-reported; cross-platform totals are indicative because platforms define reach/views differently. **Conversion metrics** are used only where legitimately measurable (e.g., volunteer sign-ups on owned forms with consent).

21.2 Content Performance Matrix

Content is analysed by format, topic/theme, length, language, publishing time, engagement, retention (avg watch %), reach, watch time, shares and comments. The matrix shows median engagement rate per format × theme cell with cell counts; cells with n < 25 are marked directional.



Content Performance Matrix — DEMO DATA

Use: **format and theme planning at aggregate level** (e.g., “short explainer videos on youth employment outperform static images by ~1.8×”). The matrix is **not** used to tailor messages to individuals or sensitive audience segments.

21.3 Experiment & test registry (added capability)

Content learning is only reliable when tests are designed up-front.

Field	Description
Hypothesis	e.g., “Hindi captions increase average watch % for 30–60 s explainers”
Design	Platform-native A/B (where offered) or matched time-slot rotation
Primary metric & MDE	e.g., avg watch %, minimum detectable effect 3 pts
Sample/budget	Pre-computed from power analysis
Stopping rule	Fixed horizon (no peeking) or sequential test with alpha spending
Result	Effect, CI, p-value; decision; learning logged
Guardrails	Only content-level variations (format, length, language, time); no targeting on sensitive attributes; all variants compliant (certified/labelled)

22. GIS / Geospatial Intelligence

Layer	Grain	Source
Constituency map	AC/PC	Official boundaries where available; community datasets with attribution for analysis; verify delimitation vintage
District/block/admin map	District/block	Official/State GIS portals where available
Historical result visualisation	AC × election	ECI statistical reports (public)
Turnout trends	AC × election	ECI statistical reports
Survey coverage	Zone	Survey metadata (counts of interviews; no respondent points)
Campaign activity coverage	Zone	Field activity aggregates
Event locations	Site (public events only)	Event records
Issue distribution	Zone/district	Field issue reports, call-centre categories, geo-attributable public mentions
Media distribution	District	Outlet geography
Operational coverage	Zone	Field + call-centre aggregates

Guardrails: no individual voter targeting; no household-level plotting; small-area suppression (e.g., hide cells with < 10 records); **maps depicting India’s national or state boundaries must use official boundaries (Survey of India) — incorrect external boundaries are a legal risk** [VERIFY exact requirements with legal].

23. Reporting

23.1 Sample Daily War-Room Report (DEMO DATA)

DAILY WAR-ROOM REPORT — Aranyapur AC (FICTIONAL) — Friday 12 September 2026, 07:00 IST *Coverage: 11 Sep 2026 (00:00–23:59). DEMO DATA — SYNTHETIC. Prepared by: AI draft + analyst review (R. Analyst). Approved by: Analytics Lead.*

1. Executive Summary

- Overall Campaign Intelligence Health: **74 / 100 (Amber)** — Field domain below target in Hill Block; all other domains green/amber.
- Social conversation volume is back within its expected range after the 31 Aug water-supply spike; **Jobs & skills** is the fastest-rising issue (+1.8 pts share vs 7-day baseline).
- Helpline answer rate (84.4%, 7-day) remains **below the 90% SLA** for the second week.

2. Major Changes (vs trailing 7-day baseline)

Signal	Yesterday	Baseline	Assessment
Public mentions (de-dup.)	1,344	1,411/day	Within range
Negative share (social)	41.2%	40.8%	No significant change
Media items	122	151/day	−19%, within range
Field completion (7-day)	83.2%	target 85%	Below target (Hill Block

Signal	Yesterday	Baseline	Assessment
			67.4%)

3. Digital Performance — ER 3.49% (+0.18 pts W/W), impressions 9.21 M (+0.9%), video views −3.8% W/W; CPM ₹116 (+4.5%). All boosted items carry certification references.

4. Social Intelligence — Top issues by share: Jobs & skills 16.3%, Water supply 14.2%, Prices 12.9%, Roads & transport 12.7%. *Coverage: public posts only; 18% geo-attributable; not representative of the electorate.*

5. Emerging Issues — Cluster “skill-centre admissions” (Hinglish, 3 days, growing 2.1× baseline, n = 240) — **Medium**; analyst validated; no misinformation-risk flags.

6. Media Intelligence — 122 items; 23.8% negative tone; top topics Roads & transport (18), Prices (17), Jobs & skills (16); regional/hyperlocal share 52.5%.

7. Survey Movement — No new wave. Latest (Wave 3, fieldwork 4 Sep, n_eff 3,704): water supply salience 21.0% ± 1.3 (significant rise from 13.0% in Wave 1).

8. Field Operations — 1,520 planned activities in last 7 days; 83.2% completed; 79.3% reports on time; bottleneck: Hill Block (Z19–Z24) completion 67.4%. 1,842 local issue reports logged.

9. Alerts — 0 Critical · 1 High (helpline SLA breach, open, owner: Call-centre Manager) · 2 Medium (Hill Block completion; X-platform connector latency) · 4 Low.

10. Data Quality — 11/14 sources green; 2 amber (X connector latency; field form late submissions); 1 red (regional TV transcripts feed delayed 26 h — vendor notified).

11. Compliance — 0 exceptions. 12 paid items live, all certified; 3 AI-assisted items, all labelled with responsible-entity disclosure; no silence period in the next 7 days for this AC.

12. Recommended Analytical Follow-ups (operational / analytical only) 1. Field Ops to review Hill Block staffing and connectivity; Data team to check whether late reports under-count completion. 2. Call-centre Manager to review peak-hour staffing; analytics to break down wait time by hour and language. 3. Research to consider adding a skills/training question to Wave 4 to test whether the social trend reflects broader salience. 4. Data team to escalate the TV-transcript feed delay with the vendor under the SLA.

Sources: metric store v1.4; social.mentions_daily; media.mentions_daily; digital.content_daily; field.activity; ops.call_centre; survey.w3. All figures DEMO DATA.

23.2 Weekly Executive Report (10–15 pages)

Page	Section	Content
1	Executive Summary	5 key findings, Health Score trend, top risks, decisions requested
2	KPI Scorecard	Level-2 domain scores and Level-3 KPIs vs targets, W/W and 4-week trend
3	Trends	Cross-source timeline of the week’s major developments
4	Digital	Scorecard, platform comparison, top/bottom content, spend pacing
5	Social	Volume, sentiment with CIs, issue shares, emerging clusters, coverage notes
6	Media	Volume/tone, story threads, regional coverage
7	Survey	Latest wave, significant changes, sample health, methodology notes
8	Field	Completion, coverage, bottlenecks, data-submission quality
9	Geography	Zone/AC maps: activity coverage, issue distribution, survey coverage
10	Forecast	Next-2-week forecasts with intervals; back-test accuracy
11	Risks	Operational, data, compliance and reputational risk items with owners
12	Data Quality	Source health, incidents, fixes
13	Compliance	Certification, AI labels, spend vs limits, upcoming deadlines
14	Next-week analytical priorities	Questions to answer, tests to run, data to acquire
15	Appendix	Definitions, methodology, sources

24. Team Structure & Operating Model

24.1 Three staffing scenarios

Role	A. POC / MVP (5–7)	B. Single constituency / by-election (8–15 core + ops support)	C. Multi-constituency / state (20–50+)
Programme Director	0.25 (shared)	0.5	1
Product Manager	1 (combined with PM)	1	1–2
Project Manager	(combined)	1	1–2
Analytics Lead	1	1	1 + 2–4 domain leads
Data Scientists	1	2	4–8
Data Analysts	1	2–3	6–12
Data Engineers	1	2	3–6
BI Developer	0.5	1	2–3
ML/NLP Engineer	1	1	2–4
GIS Analyst	—	0.5	1–2
Social Listening Analyst	(DA covers)	1	2–4 (by language)
Media Intelligence Analyst	—	1	2–4 (by language/region)
QA / Data Operations	—	1	2–4 (24×7 rota)
Security / Privacy Lead	0.25	0.5	1
Political/Communications Liaison (client-side or agency)	Client-nominated	1 (client/agency)	1–3 (client/agency)
Core total	**5–7 FTE**	**~12–15 FTE**	**~32–60 FTE**
Operations support (not core)	—	Survey vendor, field-form ops, call-centre (client)	Plus 24×7 war-room shifts, language annotators

24.2 Role cards

Role	Responsibilities	Required skills	Daily outputs	Tools	Experience
Program Director	Accountable for outcomes, client relationship, risk, commercial	Programme leadership, analytics delivery, stakeholder management, ethics judgement	Leadership stand-up, risk log review, escalation decisions	Roadmap, RAID log	15+ yrs
Product Manager	Owns requirements, backlog, UX, acceptance	Product discovery, analytics products, UX	Prioritised backlog, user feedback synthesis	Jira/Linear, Figma	8+ yrs
Project Manager	Plan, dependencies, status, vendor coordination	Delivery management, agile	Status report, dependency tracker	Jira, MS Project	6+ yrs
Analytics Lead	Analytical quality, KPI framework, report sign-off	Statistics, storytelling, domain knowledge	Approved daily report, KPI changes	SQL, Python, BI	10+ yrs
Data Scientist	Statistics, survey analytics, forecasting, NLP, model validation, AI analytics	See §24.3	Analyses, model runs, insight notes	Python, SQL, statsmodels, sklearn, Transformers	4–10 yrs
Data Analyst	Dashboards QA, ad-hoc analysis, report drafting, data checks	SQL, BI, Excel, domain	Report sections, analysis tickets	SQL, Power BI, Python	2–6 yrs
Data Engineer	Connectors, pipelines, warehouse, quality tests, lineage	Python, SQL, dbt, orchestration, cloud	Pipeline health, new sources, fixes	Dagster/Airflow, dbt, Postgres/BigQuery	4–8 yrs
BI Developer	Dashboards, semantic model, RLS, performance	Power BI/React, DAX/TS, data viz	Dashboard releases	Power BI / Next.js	4–8 yrs
ML/NLP Engineer	Multilingual models, embeddings, RAG, serving, monitoring	Transformers, Indic NLP, MLOps	Model releases, eval reports	PyTorch, HF, FAISS/pgvector, FastAPI	4–8 yrs

Role	Responsibilities	Required skills	Daily outputs	Tools	Experience
GIS Analyst	Boundaries, geocoding (aggregate), map layers	GIS, spatial SQL, cartography	Map layer updates	QGIS, PostGIS	3–6 yrs
Social Listening Analyst	Monitor public conversation, validate clusters, annotate	Languages, platforms, critical reading	Social section of report, validated alerts	Listening tool, Command Center	2–5 yrs
Media Intelligence Analyst	Monitor media, story threads, summaries	Languages, media landscape	Media section, story log	Media monitoring feed	2–5 yrs
QA / Data Operations	Data checks, pipeline monitoring, 24×7 triage	SQL, runbooks	Data-quality log, incident tickets	Monitoring, ticketing	2–5 yrs
Security / Privacy Lead	Security architecture, DPDP alignment, access reviews, incidents	Cloud security, privacy law familiarity	Access review, security findings	IdP, SIEM, KMS	8+ yrs
Political / Communications Liaison	Translates leadership questions into analytical asks; context for interpretation; ensures ethical scope	Campaign communications, ethics, domain	Question backlog, context notes	Command Center, AI Analyst	8+ yrs

24.3 The Data Scientist role — in detail

> **"A Data Scientist is not just someone who builds ML models; the role is to convert data into statistically defensible decision-support."**

#	Contribution	What it looks like in this programme
1	Statistical analysis	Period comparisons with proper tests; effect sizes rather than raw differences
2	Survey analysis	Weighting (raking), design effects, MoE, significance, cross-tab suppression
3	Trend detection	Rolling baselines, change-point detection, seasonality adjustment
4	Forecasting	Issue/media/digital/ops forecasts with intervals; back-testing
5	Anomaly detection	Robust thresholds, false-positive tuning with war-room feedback
6	NLP	Taxonomy design with analysts; model selection; code-mixed text handling
7	Sentiment analysis	Gold-set design, per-language validation, calibration
8	Topic modelling	Embedding clustering, coherence checks, analyst-in-the-loop naming
9	Content-performance analysis	Format × theme matrix, driver regression, confounder checks
10	Experiment analysis	Power analysis, pre-registration, correct stopping rules
11	Attribution / measurement (where valid)	Only with experimental or quasi-experimental designs; no over-claiming from last-click data
12	KPI modelling	Metric definitions, composite index design and sensitivity analysis
13	Model validation	Hold-outs, rolling-origin evaluation, calibration
14	Uncertainty estimation	Bootstrap, analytical CIs, prediction intervals
15	Data-quality analysis	Quantifying coverage/duplication; impact of quality issues on conclusions
16	Executive insights	Plain-language "what changed / how sure are we / what to check next"
17	AI-agent development	Tool definitions for the AI Analyst; guardrails; evaluation sets
18	RAG analytics	Retrieval quality (recall@k, citation validity), chunking strategies
19	Automated reporting	Parameterised report generation from the metric store
20	Model monitoring	Drift, performance decay, alert precision
21	Bias checks	Per-language/per-region performance; survey representativeness
22	Reproducibility	Version-controlled code, pinned environments, data snapshots, seeds
23	Documentation	Model cards, analysis notes, methodology appendices

A day in the life (campaign phase): 06:30 pipeline and alert review → 07:00 daily report sign-off with Analytics Lead → 09:00 war-room stand-up → deep-dive analysis on an open question (e.g., "is the helpline dip driven by language mix?") → model monitoring check → 16:00 insight note to leadership → evening: gold-set review with annotators.

24.4 Operating rhythm

Cadence	Forum	Output
Daily 07:00	Daily War-Room Report published	Report + alerts
Daily 09:00	War-room stand-up (15 min)	Actions and owners
Twice weekly	Data-quality & model review	Fixes, recalibrations
Weekly	Executive review (45 min)	Weekly report, decisions
Fortnightly	Governance board (data, privacy, compliance)	Approvals of new sources/purposes/models
Monthly	Commercial & roadmap review	Scope changes, cost report

24.5 SLAs (indicative — Professional tier)

Service	SLA
Daily report	Published by 08:00 IST, 7 days/week in campaign phase
Critical alert acknowledgement	≤ 15 min (24×7 in high-intensity phase; business hours otherwise)
Dashboard availability	99.5% monthly
Priority source freshness	≤ 60 min where source permits
Ad-hoc analysis request	Standard: 2 business days; urgent: same day (capped per week)
Incident response (P1)	Engage ≤ 30 min; status updates hourly

24.6 Escalation matrix

Level	Trigger	Who	Time
L1	Alert raised / data issue	On-duty analyst / Data Ops	Immediate
L2	High alert unresolved 1 h; pipeline down > 1 h; client-visible error	Analytics Lead / Data Engineering Lead	≤ 1 h
L3	Critical alert; compliance exception; security incident; SLA breach	Programme Director + Security/Privacy Lead + Client Compliance Officer	≤ 30 min
L4	Legal/regulatory exposure; data breach; major reputational risk	Agency leadership + Campaign legal counsel	Immediate

24.7 Business continuity (added capability)

Poll-week runbooks cover: cloud-region incident (warm standby), connector/API outage (manual file-drop fallback), LLM provider outage (AI Analyst degrades to dashboards + templated summaries), key-person absence (cross-trained pairs), power/network failure at war-room (remote access with MFA, mobile hotspots), and vendor feed delays (SLA escalation and coverage banners).

25. Proof of Concept (4–6 weeks)

25.1 Plan

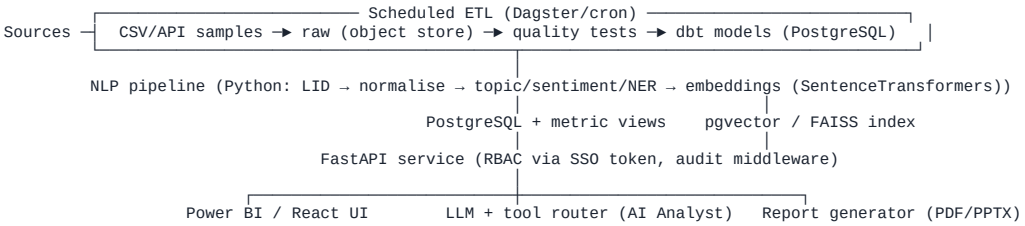
Week	Focus	Deliverables
1	Data discovery · source mapping · architecture · KPI definition	Source Registry v1; data-access status per platform (verified); architecture decision record; KPI tree & metric definitions v1; RBAC matrix; POC success criteria signed
2	Data ingestion · cleaning · database · first dashboard	Pipelines for 6+ sources (synthetic + authorised samples); quality tests; PostgreSQL model; Executive Overview v1
3	Social/NLP pipeline · survey analytics · historical analytics	Language ID, topic, sentiment models with gold-set metrics; survey engine (weights, MoE, significance); ECI historical results layer
4	AI Analyst · alerts · automated reporting	AI Analyst with tools + RAG + citations; alert engine (4 levels); auto-drafted daily report
5	Validation · security · UX improvements	Model validation report; hallucination/red-team test results; SSO/MFA/RBAC; audit logs; usability fixes
6	Executive demo · documentation · handover	Live demo; model cards; data dictionary; runbooks; production recommendation & plan

A 4-week “fast” variant compresses weeks 5–6 into weeks 3–4 by limiting scope to 4 modules.

25.2 Demo scenario — fictional constituency

- **Aranyapur AC (FICTIONAL)**: 4 blocks (North, River, Market, Hill), 24 zones (Z01–Z24), six-week window (1 Aug–11 Sep 2026).
- **100,000 synthetic records** (no real voter database): 60,000 public-style social mentions; 6,000 media items; 12,000 anonymous survey responses in 3 waves; 9,000 field activities; 1,500 owned content posts with performance and spend; 11,500 call-centre operational records.
- **Embedded storyline** for the demo: a synthetic water-pipeline outage on 31 Aug produces a social spike, a regional media surge, a helpline service dip and a significant rise in survey salience — showing how the Command Center links signals across sources.
- Every screen and export carries the label **“DEMO DATA — SYNTHETIC”**. Generator: `demo/generate_synthetic_data.py` (seeded, reproducible); charts: `demo/build_charts.py`.

25.3 Technology POC architecture



Conceptual API endpoints (full sketch in `poc/openapi.yaml`):

Method	Endpoint	Returns
GET	/dashboard/overview	Health score, domain scores, KPI strip, top changes, data-as-of
GET	/social/trends?from&to&topic&platform&language	Daily volume, anomaly flags, sentiment with CIs, coverage index
GET	/media/intelligence?from&to	Volume by outlet type, tone, issues, story threads
GET	/survey/summary?wave&question	Weighted estimates, MoE, n, n_eff, significance vs previous wave
GET	/field/operations?block&zone	Completion, coverage, on-time, bottlenecks
GET	/digital/scorecard?period	Scorecard metrics vs prior period and targets
GET	/alerts?severity&status	Alert queue
POST	/alerts/{id}/ack	Acknowledge/escalate/close with note
GET	/compliance/items?status	Content items with approval/certification/label status
GET	/data-quality/sources	Source registry health
GET	/metrics/{metric_id}	Metric definition + values (metric store)
POST	/ai/query	Answer with facts, interpretation, citations, data-as-of, confidence
GET	/reports/daily/{date}	Rendered daily report
GET	/lineage/{metric_id}	Source-to-metric lineage

25.4 POC acceptance criteria (measurable)

#	Criterion	Test	Pass threshold
AC-01	Dashboard loads successfully	Scripted load of 6 POC modules, 20 runs	P95 < 3 s; 0 errors
AC-02	Data refresh works	Scheduled ETL over 5 consecutive days	≥ 99% job success; freshness SLA met
AC-03	Source lineage available	Random 10 KPI tiles traced to source	10/10 traceable
AC-04	NLP pipeline works end-to-end	Process 50k texts	≥ 99% processed; throughput documented
AC-05	Sentiment validated on test set	Held-out gold set (≥ 500/lang for Hindi, English, Hinglish)	Macro-F1 ≥ 0.70 per language; report per class

#	Criterion	Test	Pass threshold
AC-06	Issue detection works	Back-test on injected/known events	Detects ≥ 90% of labelled events; ≤ 6 h lag; precision ≥ 60%
AC-07	AI Analyst answers only from approved data	60-question eval set incl. 15 out-of-scope/red-team	≥ 95% grounded answers; 100% correct refusals on policy-violating prompts; 0 fabricated numbers
AC-08	Citations displayed	Same eval set	100% of factual answers carry ≥ 1 valid citation and data-as-of
AC-09	Role-based access works	Test users for 4 roles	0 unauthorised data exposures in 40 test cases
AC-10	Audit logs work	Logins, queries, exports, AI prompts, approvals	100% of test events captured, immutable
AC-11	Report generation automated	Daily report for 5 days	Auto-draft ≥ 70% of sections; analyst edit time ≤ 30 min
AC-12	Alerts trigger correctly	Synthetic scenarios (spike, pipeline failure, missing AI label, spend threshold)	100% trigger at correct severity; ≤ 5 min latency
AC-13	Model metrics documented	Model registry	Model card for every POC model

POC success = all Must criteria (AC-01–AC-10, AC-12) met and ≥ 2 of 3 remaining; leadership usability score ≥ 4/5.

26. Roadmap

26.1 30 / 60 / 90 days (post-POC production)

Horizon	Theme	Outcomes
Day 0–30	**Foundation**	Production cloud (Indian region), SSO/MFA/RBAC, Source Registry, 10–15 sources live, metric store v1, Executive Overview + Digital + Social + Field modules, daily report automated, governance board running
Day 31–60	**Intelligence**	Media, Survey, Issue, Geographic modules; multilingual NLP expanded to target regional languages; alert engine tuned; experiment registry; compliance control tower v1
Day 61–90	**AI + Automation + Command Center**	AI Analyst GA with evaluation harness; RAG library; forecasting module; weekly report automation; war-room screens; DR test; pen-test; adoption KPIs

26.2 Election-cycle roadmap

Phase	Analytics focus	Compliance & operations focus
Pre-campaign (months before notification)	Baselines: historical results, public indicators, issue baselines, digital benchmarks; survey Wave 0; taxonomy & gold sets	Source authorisations, DPAs, consent notices, RBAC, policies, training
Campaign (after announcement/MCC)	Daily reports, alerts, content learning, field coverage, media monitoring	MCC-aware workflows; MCMC certification tracking; AI-label ledger; spend tracking
High-intensity phase (final weeks)	Intraday refresh on priority sources; 24×7 war-room; tighter alert SLAs	Takedown SLA timers; heightened security monitoring; DR warm standby
Final phase (48-hour silence window per phase)	Internal monitoring continues; **no publishing workflows**; operational analytics only	Silence-period calendar guard active; paid items verified paused; exit-poll window restrictions
Poll day	Operational analytics only: helpline service levels, field logistics status, incident logs, media/social monitoring for misinformation-risk (human-verified, routed to legal/comms and official channels)	Full compliance; no election-matter publication in restricted windows; incident escalation matrix at L3/L4 readiness
Post-poll	Measurement: campaign operations review, content learnings, forecast accuracy review, data-quality retrospective	Expenditure statement support (digital spend reconciliation), record retention, data deletion per schedule, lessons-learned report

27. ROI / Value Framework

We do **not** claim “we will guarantee victory”. We measure what the platform can credibly change: speed, quality, visibility and learning.

Value driver	KPI	Baseline method (Week 1)	Target (Day 90)
Reporting time	Analyst hours to produce daily + weekly reports	Time-and-motion on current process	−60%
Manual work	% of recurring report content auto-generated	Report audit	≥ 70%
Issue detection speed	Median lag onset → alert (back-tested on known past incidents)	Reconstruct past incidents	≤ 2 h for priority sources
Data quality	Sources with automated tests; quality score	Source audit	100%; avg score ≥ 85
Decision cycle	Time from signal to assigned owner/action	War-room log	≤ 4 h for High/Critical
Dashboard adoption	Weekly active named users; AI Analyst queries/user	—	≥ 80% WAU
Content learning	Documented tests with valid results; learnings applied	Experiment registry	≥ 2 valid tests/week (Professional)
Operations visibility	% of planned field activities with structured reports	Field audit	≥ 95%
Media coverage	% of priority outlets/languages monitored	Outlet list	≥ 90%
Model accuracy	Per-language F1; forecast MAPE; interval coverage	Gold sets / back-tests	Meets model-card thresholds
Report automation	Automated reports on schedule	—	100% on-time
Compliance assurance	Paid items with certification ref; AI items labelled	Compliance audit	100%

Illustrative efficiency value (not a revenue claim): if 6 analysts currently spend ~3 hours/day compiling reports, a 60% reduction releases ~11 analyst-hours/day for analysis — roughly 1.4 FTE redeployed to higher-value work.

28. Commercial Model

> **Indicative only; the final commercial proposal depends on constituency, election timeline, data availability, staffing, SLA and technology requirements.** All figures in INR, exclusive of GST; third-party pass-through at cost plus a disclosed handling fee.

28.1 Cost components

#	Component	What it covers	Pricing basis
1	**One-time setup**	Discovery, architecture, cloud landing zone, connectors, data model, metric store, dashboards, security set-up, training	Fixed fee by tier
2	**POC**	4–6 week proof of concept (§25)	Fixed fee; 50% creditable against setup if contracted within 30 days
3	**Monthly analytics retainer**	Named team, operating rhythm, daily/weekly reports, analysis, model upkeep	Monthly by tier / FTE mix
4	**Technology/ platform fee**	Platform licence/IP use, maintenance, upgrades, support	Monthly
5	**Cloud/API**	Cloud compute/storage, LLM tokens, paid APIs (e.g., X API credits), map services	Pass-through + handling fee; monthly budget caps
6	**Survey/data vendor**	CATI/F2F/online survey waves; social-listening and media-monitoring licences; broadcast transcripts	Vendor quotes, passed through
7	**Field operations enablement**	Form app configuration, training, field data-ops support (field staff themselves are client-side unless separately scoped)	Setup + monthly
8	**24×7 war-room support**	Shift-based analysts, on-call engineering in high-intensity phase	Per pod per month

#	Component	What it covers	Pricing basis
9	**Custom AI development**	New models, agents, integrations beyond tier scope	Per sprint or rate card
10	**Travel / on-site support**	War-room presence, workshops	At actuals, pre-approved

28.2 Three commercial tiers (indicative bands)

	Starter	Professional	Enterprise
Typical scope	1 AC or by-election; 8–12 weeks	1 PC or up to ~10 ACs; 3–6 months	Multi-constituency / state; 6–12 months
Modules	Executive, Digital, Social, Field, Alerts, Daily report, AI Analyst (basic)	All 14 modules; compliance tower; forecasting; experiments	All + custom models, multi-language expansion, multi-tenant views
Team (core FTE)	6–8	12–15	30–60
Support	Business hours + on-call for Critical	Extended hours; 24×7 in final 2 weeks	24×7 war-room throughout campaign phase
1. One-time setup	₹15–30 lakh	₹40–80 lakh	₹1.2–3.0 crore
2. POC (if taken)	₹18–35 lakh	₹18–35 lakh	₹25–45 lakh (broader discovery)
3. Monthly analytics retainer	₹12–22 lakh	₹30–50 lakh	₹0.8–2.0 crore
4. Technology/platform fee (monthly)	₹1.5–3 lakh	₹4–8 lakh	₹10–25 lakh
5. Cloud/API (monthly, pass-through est.)	₹0.5–1.5 lakh	₹2–6 lakh	₹8–30 lakh
6. Survey/data vendor	Per vendor quote (see 28.3)	Per quote	Per quote
7. Field ops enablement	₹2–4 lakh setup	₹4–8 lakh setup + ₹1–2 lakh/month	Scoped
8. 24×7 war-room (per pod/month)	Optional ₹8–12 lakh	₹10–15 lakh	Included (sized by pods)
9. Custom AI development	Rate card	₹5–9 lakh per 2-week sprint	Dedicated pod
10. Travel/on-site	At actuals	At actuals	At actuals

28.3 Third-party cost guidance (to be validated with vendor quotes)

Item	Indicative band	Note
CATI survey	~₹250–600 per complete	Length, language, incidence dependent **[VERIFY]**
Face-to-face (tablet) survey	~₹450–1,200 per complete	Geography and sampling design dependent **[VERIFY]**
Social-listening licence	~₹1–10 lakh/month	Volume, platforms, languages
Media monitoring (print/online)	~₹1–5 lakh/month	Outlets and languages
Broadcast transcripts (licensed)	~₹2–8 lakh/month	Channels and hours
LLM usage	Capped per user/day; typically a small share of cloud/API	Token budgets enforced
X API	Pay-per-use credits (reported since Feb 2026)	Budget cap and read limits required

28.4 Rate card (for change requests; per person-day, indicative)

Role	₹/day
Programme Director	35,000–50,000
Analytics Lead / Security & Privacy Lead	20,000–28,000
Senior Data Scientist / ML-NLP Engineer	16,000–25,000

Role	₹/day
Data Scientist / Data Engineer	12,000–18,000
Product / Project Manager	12,000–18,000
BI Developer / GIS Analyst	10,000–14,000
Data Analyst	7,000–11,000
Social / Media Intelligence Analyst	6,000–9,000
QA / Data Operations	5,000–8,000

28.5 Worked example (Professional, 4 months, indicative)

Line	Calculation	Indicative
Setup	one-time	₹60 lakh
Retainer	4 × ₹40 lakh	₹1.60 crore
Platform fee	4 × ₹6 lakh	₹24 lakh
Cloud/API	4 × ₹4 lakh	₹16 lakh
Survey	2 waves × ~₹20 lakh	₹40 lakh
War-room pod	1 month	₹12 lakh
Travel	estimate	₹5 lakh
Total		**≈ ₹3.17 crore** (range roughly ₹2.3–4.5 crore)

28.6 Cost guardrails (added capability)

Monthly cost report by component; per-module cloud budgets with alerts at 80%/100%; LLM token caps per user/day; approval required for new paid data sources; pass-through invoices shared transparently.

29. Client Dependency Checklist

#	Dependency	Needed by	Owner (client/agency)
1	Campaign scope (objectives, questions leadership wants answered)	Week 1	Agency lead
2	Geography (states, PCs/ACs, sub-units)	Week 1	Agency lead
3	Timeline (key dates, expected phases)	Week 1	Agency lead
4	Existing data inventory and samples	Week 1	Team leads
5	API access / admin authorisation for owned accounts	Week 1–2	Digital lead
6	Dashboard / BI tool access (if reusing)	Week 1	IT
7	Survey data, questionnaires, methodology, weights	Week 2	Research lead
8	Authorised first-party data with lawful-basis evidence	Week 2	Data owner + legal
9	Field report formats and access to the field app	Week 2	Field head
10	Media-monitoring access/licences	Week 2	Media cell
11	Compliance contact (named)	Week 1	Agency
12	Legal contact (named)	Week 1	Campaign
13	Cloud preference and accounts	Week 1	IT
14	Security requirements (IdP, device policy, residency)	Week 1	IT/security
15	Reporting cadence and distribution list	Week 1	Leadership office
16	Decision-makers and approval matrix	Week 1	Leadership office
17	SLA expectations and support hours	Week 1	Agency lead
18	Issue taxonomy input and language priorities	Week 1–2	Research + comms
19	Annotators / language reviewers (or approval to use Partner annotators)	Week 2	Agency
20	Brand assets **with authorisation** (if any official marks are to be used)	Before external use	Agency

30. Risk Register

Scale: Probability and Impact 1 (low) – 5 (high). Score = P × I.

#	Risk	P	I	Score	Mitigation	Owner
R1	Poor data quality (late, incomplete, inconsistent)	4	4	16	Quality gates, source scorecards, badges on tiles, data-ops rota	Data Eng Lead
R2	Incomplete or unavailable APIs	4	3	12	Week-1 verification; file-drop fallbacks; licensed vendors	Data Eng Lead
R3	Platform restrictions / policy changes	3	4	12	Vendor diversity; contractual review; monitor developer changelogs	Product Manager
R4	Biased surveys (sampling, non-response, mode)	3	4	12	Design review, weighting, QC, benchmark checks, AAPOR-style disclosure	Research Lead / DS
R5	Social-media non-representativeness	5	3	15	Coverage index; mandatory caveats; triangulation with survey/field	Analytics Lead
R6	Model drift	3	3	9	Monitoring, monthly re-validation, gold-set refresh	ML Lead
R7	LLM hallucination	3	5	15	Tool-first answers, numeric/citation checks, eval harness, refusal policy	ML Lead
R8	Overfitting (short histories)	3	3	9	Simple baselines, rolling-origin validation, parsimony	DS Lead
R9	False-positive alerts (alert fatigue)	4	2	8	Thresholds tuned with feedback; grouping; precision KPI	Analytics Lead
R10	False-negative alerts (missed issues)	3	4	12	Multi-signal rules; back-tests; analyst scanning	Analytics Lead
R11	Compliance risk (uncertified ads, unlabelled AI, silence period)	3	5	15	Control tower rules; blocking states; legal sign-off	Compliance Officer
R12	Privacy risk (excess personal data, purpose creep)	3	5	15	Minimisation, permitted-use tags, DPIA-style review, audits	Privacy Lead
R13	Cyber risk (breach, credential theft, insider)	3	5	15	SSO/MFA, least privilege, logging, pen-test, DLP, IR runbook	Security Lead
R14	Unrealistic client expectations	4	4	16	“What we will not promise” in contract; acceptance criteria; education	Programme Director
R15	Vendor/API cost overrun	3	3	9	Budget caps; monthly cost report	Project Manager
R16	Key-person dependency	3	3	9	Cross-training; documentation; bench	Programme Director
R17	Reputational risk from misuse of outputs	2	5	10	Scope guardrails; refusal policy; audit logs; ethics escalation	Programme Director
R18	Regulatory change mid-campaign	3	4	12	Legal watch; configurable rules engine	Compliance Officer

Heat view:

Impact ↓ / Probability →	1–2	3	4–5
5	R17	R7, R11, R12, R13	—
4	—	R3, R4, R10, R18	R1, R14
3	—	R6, R8, R15, R16	R2, R5
2	—	—	R9

31. What We Should Not Promise

We will not promise	Why
Guaranteed election outcome	Outcomes depend on factors far outside analytics
Exact vote prediction	Not statistically defensible; restricted publication windows; not in scope
Complete voter-preference prediction	Neither possible nor ethical at individual level
Individual political-belief inference	Prohibited by design
Unauthorised personal-data access	Unlawful
Secret tracking	Unlawful and unethical
Illegal scraping	Breaches law and platform terms
Private-message monitoring	Unlawful; privacy violation
Guaranteed viral content	Platform algorithms and audiences are not controllable
Guaranteed persuasion	Out of scope; manipulative framing rejected
Guaranteed ROI	We commit to measurable operational KPIs (§27), not outcomes

32. Visual Design & Asset Sourcing

32.1 Design direction

A premium political-technology aesthetic that is **professional, not propagandistic**: deep navy/charcoal as the dominant field; saffron as the signal/accent colour; white space; subtle gold for highlights; clean data visualisation; abstract network/"constellation" motifs representing connected data; dashboard mock-ups; no inflammatory imagery, no party symbols, no leaders' photographs.

Token	Hex	Use
Deep navy	#0B1F3A	Title/closing slides, headers
Charcoal ink	#1B2330	Body text
Saffron	#E2711D	Accent, series 1, key numbers
Signal blue	#2F6DB5	Series 2
Teal	#1A9A8F	Series 3
Ochre-gold	#A67C1A / #C9A227	Series 4 / highlights
Light surface	#F4F6F9	Cards

The categorical palette (saffron, blue, teal, ochre) passed an automated colour-vision-deficiency and contrast validation on a light surface.

Branding rule: party marks are used **only where the Agency holds written authorisation**; the deck contains a clearly labelled **"[Client logo placeholder — official assets only with authorisation]"** box.

Map rule: no India outline is used in the deck; any national/state map in client-facing material must use official Survey of India boundaries.

32.2 Visual assets register

Asset	Source	Licence / usage consideration	Where used
Abstract network cover visual	Generated for this project (demo/build_charts.py)	Own work; no third-party rights	Slides 1, 34
All charts (anomaly, sentiment, survey, content matrix, hex map, forecast, media, call-centre)	Generated from synthetic DEMO data	Own work; labelled DEMO DATA	Slides 12–18 and PRD
Diagrams, dashboard mock-ups, org charts, roadmap	Drawn natively in PowerPoint	Own work; editable	Slides 2–35
Optional: night-lights Earth imagery	NASA Earth Observatory (Black Marble)	NASA imagery is generally not copyrighted but NASA media guidelines apply (no endorsement implied) — **verify per image**	Optional replacement for cover
Optional: control-room / team	Unsplash / Pexels	Free commercial use under their	Optional for

Asset	Source	Licence / usage consideration	Where used
photographs		licences; avoid identifiable people without releases; no implied endorsement	team slide
Optional: icons	Google Material Symbols (Apache 2.0)	Attribution in notes	Optional
Optional: official maps	Survey of India	Official boundaries; terms per SoI	Any national/state map

No external images were used in the current deck; if suitable images cannot be safely sourced, the generated abstract visuals remain.

33. Presentation — Slide-by-Slide Content

The executive deck (`deliverables/Election_Intelligence_Command_Center_Executive_Deck.pptx`, 35 slides) follows this structure:

#	Slide	Purpose	Key content	Visual
1	Cover	Set positioning	Title, subtitle, “AI + Data + Digital Intelligence Partner”, placeholder logos	Generated network visual
2	Executive Vision	The idea in one loop	Data → Intelligence → Insight → Decision Support → Measurement → Learning	6-node loop
3	Why Campaign Intelligence Matters	Urgency	4 regulatory/operational shifts of 2025–26	Stat cards with dates
4	The Current Problem	Pain	Fragmented workflows; 6 blind spots	Current-state diagram
5	Our Solution	Before/after	Current vs future state	Two-column comparison
6	Command Center Concept	Product	9 capability pillars on one intelligence layer	Layered pillars
7	Data Ecosystem	Breadth with discipline	Six source families A–F	6 cards
8	Data Sources & Access Reality	Honesty on APIs	Platform-by-platform access status to verify	Status table
9	Data Architecture	Engineering	Sources → … → Command Center	Pipeline diagram
10	Data Model & Analytics Stack	Structure	Conceptual ERD + 4 analytics levels	ERD + ladder
11	AI / NLP Layer	Multilingual intelligence	Listening → NLP → Topic → Sentiment → Trend → Alert	Flow + language chips
12	Social Intelligence	Demo	Anomaly chart + coverage caveat	Chart
13	Media Intelligence	Demo	Weekly outlet mix + modules	Chart
14	Survey Intelligence	Rigour	Weighted CIs; methodology controls	Chart
15	Digital Analytics	Scorecard	Scorecard + content matrix	Chart + KPI cards
16	Field & Call-centre Operations	Operations	KPIs + service levels	Chart + cards
17	GIS Intelligence	Geography	Zone map + layers + guardrails	Hex map
18	Forecasting & ML	Honest prediction	Forecast with intervals + model card	Chart + card
19	AI Analyst	Grounded AI	Architecture + sample answer with citations	Diagram + chat mock
20	Real-Time Alerts	Early warning	4 levels, SLAs, examples	Severity ladder
21	Executive Dashboard	The product	Health score + domain chips + KPI strip	Dashboard mock-up
22	Compliance Control Tower	Auditability	Status table + regulatory anchors	Tower mock-up
23	Security & Governance	Trust	Controls grid	Shield grid
24	Responsible AI	Ethics	Principles + three kinds of AI	Three-column
25	Team Structure	Delivery	Three scenarios	Org chart
26	The Data Scientist Role	Depth	23 contributions in 5 clusters	Cluster cards
27	POC Plan	Next step	6-week plan + acceptance	Gantt

#	Slide	Purpose	Key content	Visual
28	Technology POC	Build	Stack + endpoints	Architecture + endpoint list
29	Roadmap	Horizon	30/60/90 + election cycle	Timeline
30	Commercial Model	Investment	Three tiers	Tier table
31	KPIs & Value	Measurable outcomes	KPI tree + value metrics	Tree + cards
32	Risks & Mitigation	Candour	Top risks heat map	Matrix
33	Boundaries & Dependencies	Expectations	What we won't promise; what we need	Two columns
34	From Campaign Data to Campaign Intelligence	Close	Platform · Team · Technology · Governance · Outcomes	Network visual
35	Sources & Notes	Credibility	Key sources, disclaimers	List

34. Critical Self-Review & Red-Team

34.1 "What important capabilities have we missed?" — checklist result

Capability	Covered in	Status
Data governance	§9	✓
Privacy	§9, §16, §17	✓
Cybersecurity	§16	✓
Data lineage	§9, §11, AC-03	✓
Data quality	§6, §9, §18 (screen 12)	✓
Model governance	§17.3	✓
AI hallucination protection	§15.5	✓
Multilingual NLP	§14.2	✓
GIS	§22	✓
Survey methodology	§8.3, §13.4	✓
Media intelligence	§13.3	✓
Digital analytics	§21	✓
Field operations	§13.5	✓
Call-centre analytics	§13.6	✓
Anomaly detection	§14 (M8), §19	✓
Forecasting	§14 (M7, M9), §17.3	✓
Experiment measurement	§21.3	✓ (added)
Automated reporting	§23	✓
Executive dashboards	§18	✓
Compliance	§10	✓
AI-content labelling	§10.2, §17.2	✓
Audit trails	§9, §10, §16	✓
Disaster recovery	§7, §16	✓
Business continuity	§24.7	✓ (added)
Vendor/API dependency	§8.4, §9, R2/R3	✓
Cost management	§28.6	✓ (added)
Staffing	§24	✓
SLA	§24.5	✓
Escalation matrix	§24.6	✓ (added)
Client dependencies	§29	✓
POC success criteria	§25.4	✓
Acceptance criteria	§25.4	✓

34.2 Red-team review

Lens	Finding	Action taken
Technical feasibility	Real-time social listening at scale depends on paid, permitted access — not free APIs	API reality table; licensed vendors; file-drop fallbacks; “verify” tags
	Hinglish/romanised text breaks off-the-shelf sentiment models	Normalisation + per-language gold sets + thresholds
	6-week POC risks over-scope	6 modules in POC; others on roadmap; 4-week variant defined
Legal/ compliance	Several ECI instructions verified through secondary reporting only	Marked [VERIFY]; legal sign-off gate in Week 1
	Exit-poll and silence-period constraints could be breached by automated publishing	Calendar guard blocks publishing workflows and restricted report types
	Using party branding without authorisation	Placeholders only; authorisation required
	Incorrect India boundaries on maps	No national outline; SoI boundaries required
Privacy	Call-centre, CRM and field data can contain personal data	Minimisation, pseudonymisation, permitted-use tags, no caller PII in POC
	Public social data still contains personal data	Aggregate early; drop handles after de-dup; no individual profiles
	“Creator amplification” could drift into profiling individuals	Restricted to public figures/media/organisations, aggregated by account type
AI hallucination	LLM may invent numbers or sources	Tool-first answers, numeric checker, system-generated citations, eval harness, AC-07/AC-08
	Prompt injection via retrieved documents	Document intake approval; injection filters; red-team suite
Data quality	Field reports late → under-counted completion	Timeliness KPI; late-report reconciliation; badges
	Survey weights unstable with small cells	Trimming; cell suppression; n_eff shown
Unrealistic claims	Health score could be misread as a win probability	Defined as operational; explicit disclaimer on screen
	Synthetic demo metrics (e.g., 2.2% MAPE) look better than reality	Model card states real-world error will be higher
Missing team roles	Annotators and language reviewers; 24×7 data-ops	Added to scenario C and dependencies
Missing costs	Listening/media licences, X API credits, annotation, pen-test, DR standby	Added to §28.3 and tiers
Missing dependencies	Legal contact, annotators, brand authorisation	Added to §29
Missing POC requirements	Usability, red-team test, RBAC test users	AC-07, AC-09, usability score

34.3 Additional capabilities added after self-review (and why)

#	Capability	Why it matters
1	Source & lawful-basis registry with permitted-use tags enforced at query time	Turns purpose limitation from policy into a technical control
2	Metric store / semantic layer	Ends “whose number is right” debates; AI and dashboards agree
3	Coverage & Representativeness Index	Prevents over-reading social data
4	Multilingual gold sets with agreement targets	Makes NLP accuracy measurable per language
5	Experiment & test registry	Turns content performance into valid learning
6	Silence-period & exit-poll calendar guard	Hard control on legally restricted windows
7	AI Content Provenance Ledger	Mirrors ECI record-keeping expectations for AI material
8	Official-boundary map rule	Avoids a common legal pitfall
9	Escalation matrix, SLAs, BCP runbooks	Poll-week resilience
10	Cost guardrails	Prevents API/LLM bill shock
11	AI red-team & hallucination harness	Evidence that the AI is safe to use
12	Exit & data-return plan	Client retains control of its data
13	Health-score “red domain cap” rule	Prevents good news hiding bad news

#	Capability	Why it matters
14	Late-report reconciliation for field data	Avoids false operational alarms

Appendix A — Sources

Retrieved/checked September 2026. Where the ECI primary document could not be retrieved directly, the reputable report of the ECI press note/advisory is cited and the item is marked [VERIFY] in the body. This appendix does not constitute legal advice.

A.1 Legal sources

ID	Source	Date	Supports
L1	ECI press note on regulation of political advertising & digital campaigning for 2026 Assembly elections — reported by ANI: https://www.aninews.in/news/national/general-news/eci-mandates-pre-certification-of-political-ads-on-social-electronic-media-ahead-of-elections20260320194334/ ; Akashvani (Prasar Bharati): https://www.newsonair.gov.in/ec-issues-order-to-regulate-political-advertisements-across-electronic-digital-media ; primary: https://www.eci.gov.in (press releases)	20 Mar 2026	MCMC pre-certification incl. social media; social-media account disclosure; 75-day expenditure statements incl. internet spend
L2	ECI on pre-certification of political advertisements in print media — Akashvani: https://www.newsonair.gov.in/election-commission-mandates-pre-certification-of-political-advertisements-in-print-media	6 Apr 2026	Print-ad pre-certification
L3	ECI advisory on labelling AI-generated/synthetic content — Business Today: https://www.businesstoday.in/india/story/eci-mandates-clear-labelling-of-ai-generated-content-in-political-campaigns-499535-2025-10-24 ; ANI: https://www.aninews.in/news/national/general-news/eci-issues-advisory-to-control-misuse-of-ai-generated-content-during-elections-urges-parties-to-report-fake-accounts-deepfakes20251025015121/ ; Akashvani: https://www.newsonair.gov.in/ec-issues-fresh-guidelines-on-ai-and-synthetic-content-use-in-bihar-poll-campaigns	24–25 Oct 2025	Label text, 10% area / first 10% audio, responsible-entity disclosure, records, 3-hour takedown
L4	ECI advisory on labelling synthetic content — Akashvani: https://www.newsonair.gov.in/eci-directs-political-parties-to-label-ai-generated-synthetic-content-in-campaigns ; IndiaAI: https://indiaai.gov.in/article/election-commission-of-india-embraces-ai-ethics-in-campaigning-advisory-on-labelling-ai-generated-content	16 Jan 2025	Earlier AI-labelling advisory
L5	ECI guidelines on responsible and ethical use of social media — DD India: https://ddindia.co.in/2024/05/eci-issues-guidelines-for-ethical-social-media-use-in-campaigns/	6 May 2024	Deepfake prohibition; 3-hour removal
L6	ECI reiteration for 2026 elections — SCC Online: https://www.scconline.com/blog/post/2026/04/21/eci-regulates-social-media-during-elections-2026/ ; MediaNama: https://www.medianama.com/2026/04/223-eci-orders-3-hour-takedown-rule-ai-fake-content-elections/	Apr 2026	3-hour takedown; labels; enforcement statistics
L7	ECI instructions on use of social media in election campaigning (25 Oct 2013) — ECI FAQ on Social Media:	2013 (FAQ updated periodically)	MCC applies online; pre-certification; social-media expenditure

ID	Source	Date	Supports
	https://old.eci.gov.in/files/file/13753-faq-on-social-media/ ; Nyaaya explainer: https://nyaaya.org/legal-explainer/social-media-advertisements/		
L8	IT (Intermediary Guidelines and Digital Media Ethics Code) Amendment Rules, 2026 — Khaitan & Co: https://www.khaitanco.com/thought-leadership/MeitY-notifies-the-IT-Amendment-Rules-2026 ; Freshfields: https://www.freshfields.com/en/our-thinking/blogs/technology-quotient/india-targets-deepfakes-and-ai-generated-content-key-changes-under-meitys-2026-102mjwn ; MeitY FAQ on SGI: https://www.meity.gov.in/static/uploads/2025/10/065b6deb585441b5ccdf8be42502a49c.pdf	Notified 10 Feb 2026; in force 20 Feb 2026	SGI labelling; 3-hour intermediary takedown
L9	Representation of the People Act, 1951 (ss. 77, 126, 126A) — India Code: https://www.indiacode.nic.in	As amended	Expenditure accounts; silence period; exit-poll restriction
L10	Report of the Committee on Section 126 of the RP Act — ECI: https://eci.gov.in/files/file/9276-report-of-the-committee-on-section-126-of-the-representation-of-the-people-act-1951/ ; PIB: https://www.pib.gov.in/newsite/PrintRelease.aspx?relid=187412	Jan 2019	Silence period and digital media; platform code of ethics (3-hour processing)
L11	ECI press release — limits of candidates' expenses enhanced: https://old.eci.gov.in/files/file/13928-limits-of-candidate%E2%80%99s-expenses-enhanced/	6 Jan 2022	₹95/75 lakh (PC), ₹40/28 lakh (AC)
L12	Digital Personal Data Protection Act, 2023 — MeitY: https://www.meity.gov.in/data-protection-framework	Aug 2023	Data-protection obligations
L13	DPDP Rules, 2025 — notified 14 Nov 2025; analysis: https://www.amsshardul.com/insight/enforcement-of-the-dpdp-act-and-notification-of-the-dpdp-rules/ ; Rule 6 text: https://www.dpdpa.com/dpdparules/rule6.html	14 Nov 2025 (phased to 14 May 2027)	Security safeguards, 1-year logs, breach reporting
L14	CERT-In Directions under s.70B(6) IT Act — https://www.cert-in.org.in ; summary: https://www.internetsociety.org/resources/doc/2022/internet-impact-brief-india-cert-in-cybersecurity-directions-2022/	28 Apr 2022 (effective 28 Jun 2022)	6-hour incident reporting; 180-day logs in India
L15	India AI Governance Guidelines — PIB/MeitY: https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/nov/doc2025115685601.pdf	5 Nov 2025	Responsible-AI principles
L16	Meta — Ads about Social Issues, Elections or Politics: https://transparency.meta.com/policies/ad-standards/SIEP-advertising/SIEP/	Current (checked Sep 2026)	Authorisation and disclaimers
L17	Google — Political content policy: https://support.google.com/adspolicy/answer/6014595 ; India election-ads pre-certificate requirements: https://support.google.com/adspolicy/contact/in_elections_pre_cert_requirements	Current (checked Sep 2026)	Verification; ECI pre-certificate per ad
L18	ECI exit-poll prohibition for 2026 Assembly	8 Apr 2026	Exit-poll window (9–29 Apr 2026)

ID	Source	Date	Supports
	elections — Newsgram: https://www.newsgram.com/india/2026/04/08/ec-restricts-exit-polls-rpa-1951		

A.2 Technical sources

ID	Source	Date	Supports
T1	Meta Graph API documentation: https://developers.facebook.com/docs/graph-api	Current	Page insights access model
T2	Instagram Platform documentation: https://developers.facebook.com/docs/instagram-platform ; Basic Display EOL (4 Dec 2024)	Current	Business/Creator-only APIs
T3	Meta Ad Library tools: https://transparency.meta.com/researchtools/ad-library-tools	Current	Ad Library API, verification
T4	Meta Content Library announcement: https://about.fb.com/news/2023/11/new-tools-to-support-independent-research/ ; CrowdTangle shutdown: https://www.cjr.org/tow_center/meta-is-getting-rid-of-crowdtangle.php	Nov 2023; 14 Aug 2024	Researcher-only access
T5	YouTube Data API v3: https://developers.google.com/youtube/v3 ; YouTube Analytics API: https://developers.google.com/youtube/analytics	Current	Public metadata; owned-channel analytics
T6	X API pay-per-use announcement: https://devcommunity.x.com/t/announcing-the-launch-of-x-api-pay-per-use-pricing/256476	2026	Pricing model (budget caps)
T7	Google Trends API (alpha): https://developers.google.com/search/blog/2025/07/trends-api	24 Jul 2025	Application-gated alpha
T8	GA4 Data API: https://developers.google.com/analytics/devguides/reporting/data/v1	Current	Website analytics
T9	Google Ads API: https://developers.google.com/google-ads/api/docs/start	Current	Owned ad-account reporting
T10	PostgreSQL: https://www.postgresql.org/docs/ ; pgvector: https://github.com/pgvector/pgvector	Current	Warehouse + vectors
T11	FAISS: https://github.com/facebookresearch/faiss	Current	Vector similarity
T12	dbt: https://docs.getdbt.com/	Current	Transformations, tests, lineage
T13	Dagster: https://docs.dagster.io/ ; Apache Airflow: https://airflow.apache.org/docs/	Current	Orchestration
T14	FastAPI: https://fastapi.tiangolo.com/	Current	API layer
T15	Sentence-Transformers: https://www.sbert.net/	Current	Embeddings
T16	Power BI row-level security: https://learn.microsoft.com/power-bi/enterprise/service-admin-rls	Current	RLS
T17	OWASP ASVS: https://owasp.org/www-project-application-security-verification-standard/	Current	App-security baseline

A.3 Data sources

ID	Source	Date	Supports
D1	ECI Statistical Reports: https://www.eci.gov.in/statistical-reports	Current	Historical results, turnout, electors
D2	PIB — Index Cards and Statistical Reports published on ECINET: https://www.pib.gov.in/PressReleasePage.aspx?PRID=2258991	2026	72-hour publication for 2026 Assembly elections
D3	TCPD Lok Dhaba (Ashoka University): https://tcpd.ashoka.edu.in/lok-dhaba/ ; https://lokdhaba.ashoka.edu.in/	Current	Cleaned historical results (licence to confirm)
D4	Census 2027 — PIB document: https://static.pib.gov.in/WriteReadData/specificdocs/docu	25 Apr 2026	Census timeline

ID	Source	Date	Supports
	ments/2026/apr/doc2026425856601.pdf		
D5	Open Government Data Platform: https://data.gov.in	Current	Public indicators
D6	DataMeet community maps — Assembly Constituencies: https://projects.datameet.org/maps/assembly-constituencies/	Current	Boundaries (CC BY 2.5 India; delimitation vintage caveats)
D7	Survey of India: https://surveyofindia.gov.in	Current	Official boundaries
D8	ECINET: https://ecinet.eci.gov.in	Current	ECI unified portal

A.4 AI/ML sources

ID	Source	Supports
A1	IndicTrans2 (AI4Bharat, MIT): https://github.com/AI4Bharat/IndicTrans2	Translation, 22 scheduled languages
A2	IndicBERT (AI4Bharat, MIT): https://huggingface.co/ai4bharat/indic-bert	Indic encoder
A3	MuRIL (Google): https://huggingface.co/google/muril-base-cased	Indic + transliterated encoder (licence to confirm)
A4	Indic NLP Catalog: https://ai4bharat.github.io/indicnlp_catalog/	Resource discovery
A5	NIST AI Risk Management Framework 1.0: https://www.nist.gov/itl/ai-risk-management-framework	AI risk governance
A6	ISO/IEC 42001:2023: https://www.iso.org/standard/81230.html	AI management systems
A7	Mitchell et al., “Model Cards for Model Reporting”: https://arxiv.org/abs/1810.03993	Model cards
A8	Gebru et al., “Datasheets for Datasets”: https://arxiv.org/abs/1803.09010	Dataset documentation
A9	Lewis et al., “Retrieval-Augmented Generation”: https://arxiv.org/abs/2005.11401	RAG
A10	Grootendorst, “BERTopic”: https://arxiv.org/abs/2203.05794	Topic modelling
A11	AAPOR Code of Professional Ethics and Practices: https://aapor.org/standards-and-ethics/	Survey transparency
A12	Kish, L. (1965). *Survey Sampling*. Wiley	Design effect, effective sample size

A.5 Design / visual sources

ID	Source	Licence / consideration
V1	Generated network cover and all charts (demo/build_charts.py)	Own work
V2	NASA Earth Observatory — Night Lights: https://earthobservatory.nasa.gov/features/NightLights ; NASA media guidelines: https://www.nasa.gov/nasa-brand-center/images-and-media/	Optional; verify per image
V3	Unsplash License: https://unsplash.com/license	Optional photography
V4	Pexels License: https://www.pexels.com/license/	Optional photography
V5	Google Material Symbols: https://fonts.google.com/icons	Apache 2.0 icons (optional)
V6	Survey of India official maps: https://surveyofindia.gov.in	Required for India boundaries

Appendix B — Glossary

Term	Meaning
AC / PC	Assembly Constituency / Parliamentary Constituency
MCC	Model Code of Conduct
MCMC	Media Certification and Monitoring Committee (district/state)
SGI	Synthetically generated information (IT Rules 2026)
DPDP	Digital Personal Data Protection Act 2023 / Rules 2025
MoE	Margin of error
deff / n_eff	Design effect / effective sample size
MAD	Median absolute deviation (robust spread)

Term	Meaning
MAPE	Mean absolute percentage error
RAG	Retrieval-augmented generation
RLS / RBAC	Row-level security / role-based access control
RPO / RTO	Recovery point / time objective
ER / CTR / CPC / CPM	Engagement rate / click-through rate / cost per click / cost per thousand impressions
FCR	First-contact resolution